

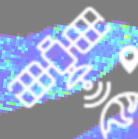
The Course

Part 9: Pol-InSAR 2

Prepared by DLR-HR's Pol-InSAR Team

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ESAMAAP

EEBI  **MASS**

Forest Height inversion from InSAR Data



$$\tilde{Y}(s_1, s_2) = \frac{\langle s_1 s_2^* \rangle}{\sqrt{\langle s_1 s_1^* \rangle \langle s_2 s_2^* \rangle}}$$

Interferometric Coherence (complex)

you measure this

$$\tilde{Y}(s_1, s_2) = Y_{\text{Sys}} Y_{\text{Tmp}} \tilde{Y}_{\text{Vol}}$$

$\tilde{Y}_{\text{Vol}}$... volume decorrelation $\tilde{Y}_{\text{Tmp}}$... temporal decorrelation

you parametrize this

$$\tilde{Y}_{\text{Vol}}(\vec{w}, \kappa_z) = e^{i\kappa_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}) e^{i\kappa_z z} dz}{\int_0^{h_v} f(z, \vec{w}) dz}$$

To isolate $\tilde{Y}_{\text{Vol}}$
1) Do something to have $f(z, \vec{w})$
2) Estimate it all $f(z, \vec{w})$

$f(z, \vec{w})$... vertical reflectivity function κ_z ... vertical wavenumber

Interferometric (volume) decorrelation is sensitive to the “visible” (forest) **height** and to the **vertical reflectivity function** $f(z, \vec{w})$ within the interferometric resolution cell.

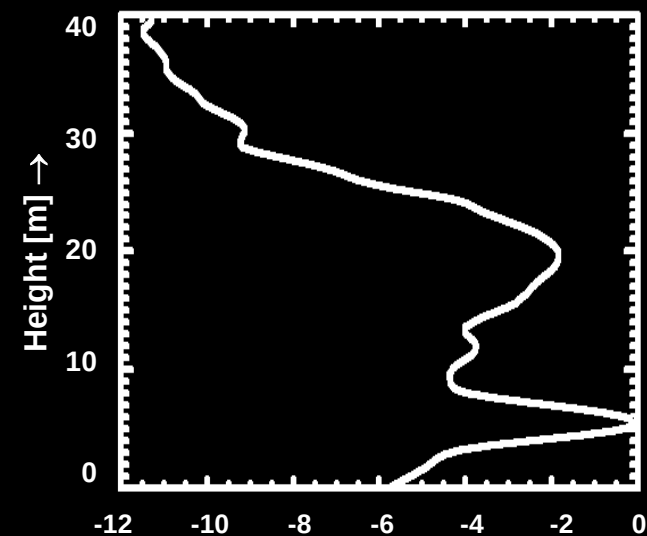


$f(z, \vec{w})$

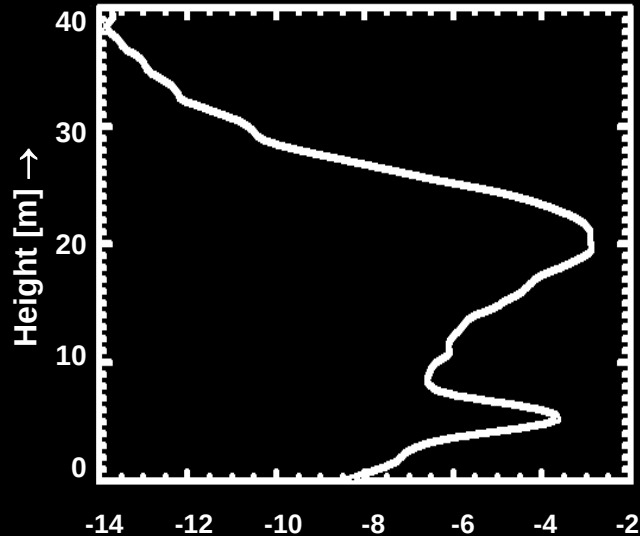
Forest Height inversion challenges:

- The parameterisation / description of $f(z, \vec{w})$
- The presence of $\tilde{Y}_{\text{Tmp}}$

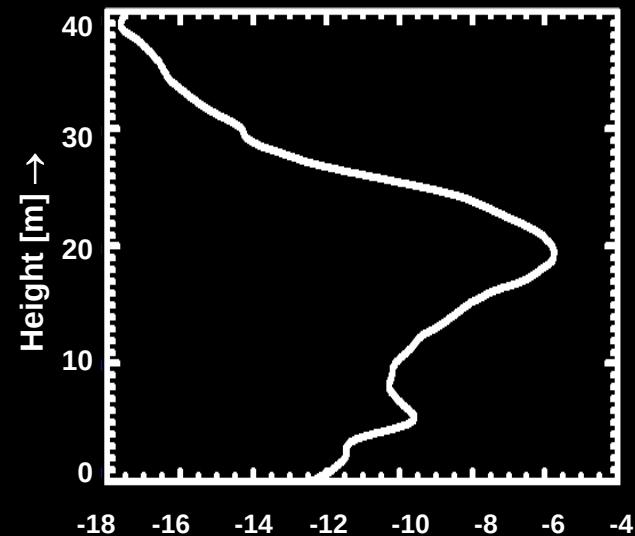
HH



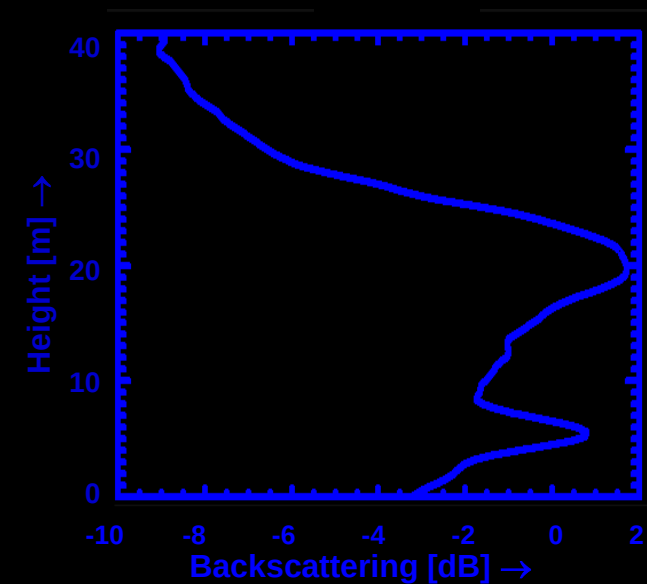
VV



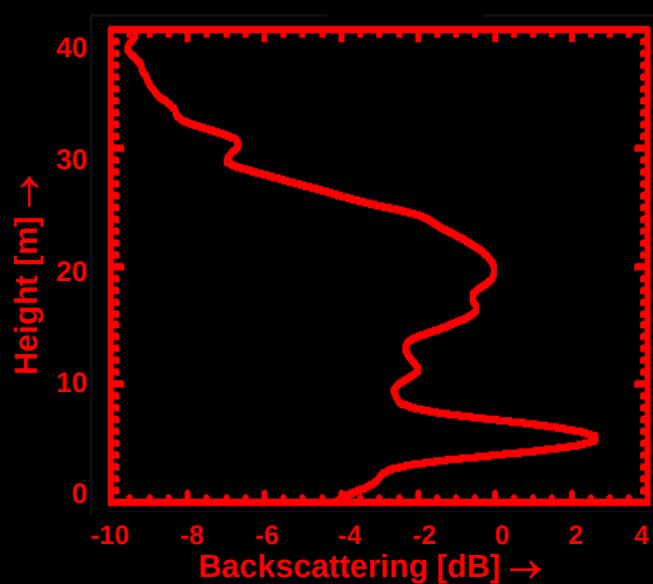
HV



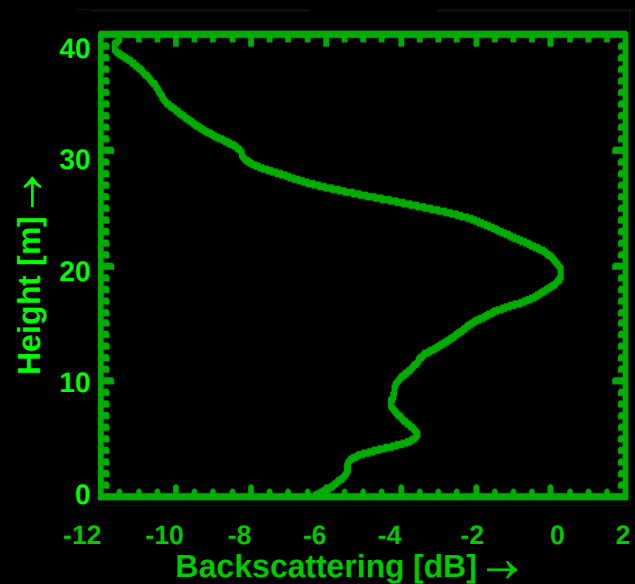
HH+VV



HH-VV



2*HV



Spruce Forest Backscattering Profiles (15-20 m height)

2 Layer Inversion Model: $f(w,z)=\text{Volume over Ground (VoG)}$



Volume Layer Ground Layer

$$f(z, \vec{w}) = m_v f_v(z, \vec{w}) + m_G(\vec{w}) \delta(z - z_0)$$

Volume Layer Coherence

$$\tilde{Y}_v = \frac{I}{I_0} \left\{ \begin{array}{l} I = \int_0^{h_v} \exp(ik_z z') f_v(z', \vec{w}) dz' \\ I_0 = \int_0^{h_v} f_v(z', \vec{w}) dz' \end{array} \right.$$

$$\tilde{Y}_{vol}(\vec{w}, \kappa_z) = \exp(i\varphi_0) \frac{\tilde{Y}_v(\vec{w}, \kappa_z) + m(\vec{w})}{1 + m(\vec{w})}$$

$$m(\vec{w}) = \frac{m_G(\vec{w})}{m_v(\vec{w}) I_0} \quad \kappa_z = \frac{\kappa \Delta \theta}{\sin(\theta_0)}$$

$f_v(z)$... volume reflectivity function

$\varphi_0 = k_z z_0$... underlying topography

Single Baseline Observations

single- / dual- / quad-pol

$$\tilde{Y}_{vol}(\vec{w}_1, \kappa_z) \quad \tilde{Y}_{vol}(\vec{w}_2, \kappa_z) \quad \tilde{Y}_{vol}(\vec{w}_3, \kappa_z)$$

1, 2, or 3 complex coherences

Total Coherence

$$\tilde{Y}(\vec{w}, \kappa_z) = Y_{Temp}(\kappa_z) \tilde{Y}_{vol}(\vec{w}, \kappa_z)$$

For a Single Baseline

3+N unknown parameters

Volume Height h_v

Topography φ_0

G/V Ratio $m(\vec{w}) = f(\text{pol})$

$f_v(z, \vec{w})$... N parameters

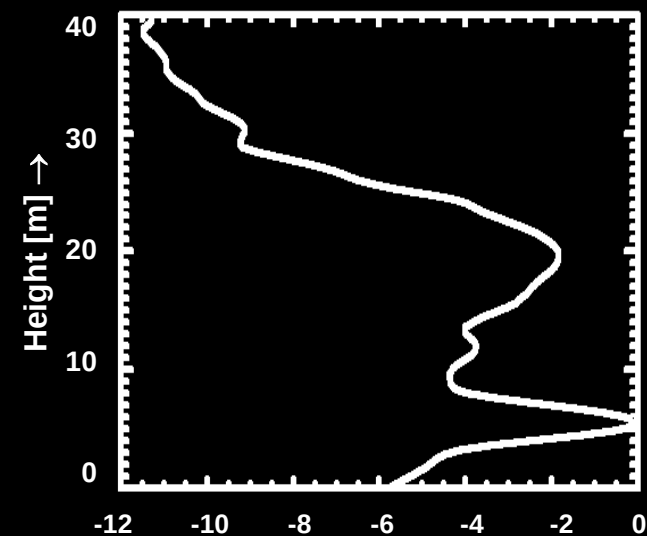
plus one more

Temporal Deco Y_{Temp}

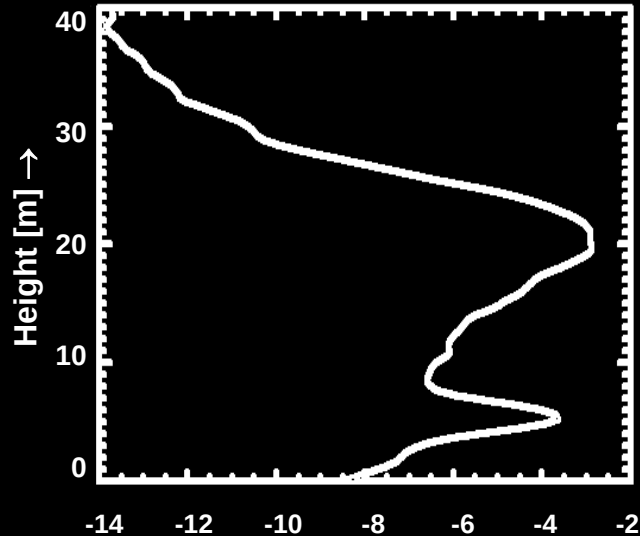
for repeat-pass implementations



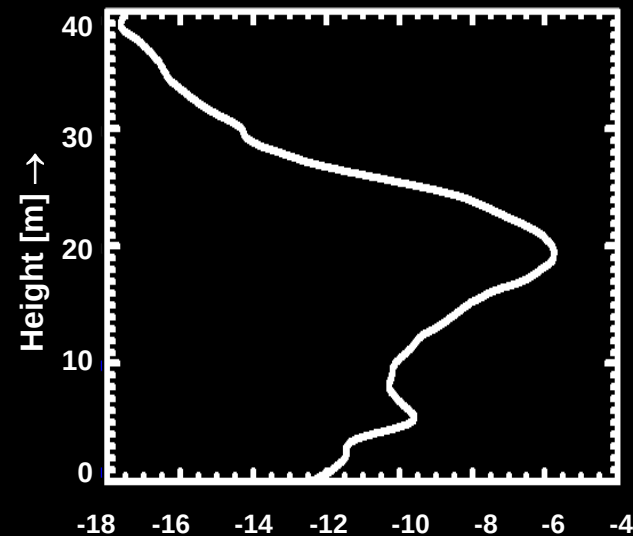
HH



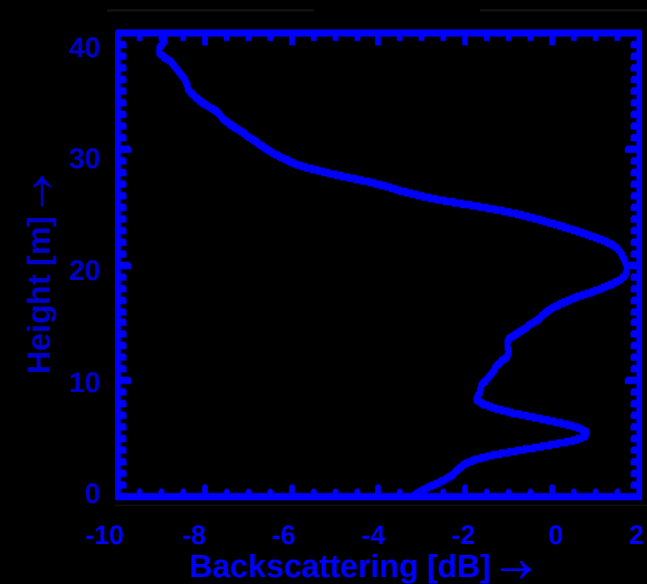
VV



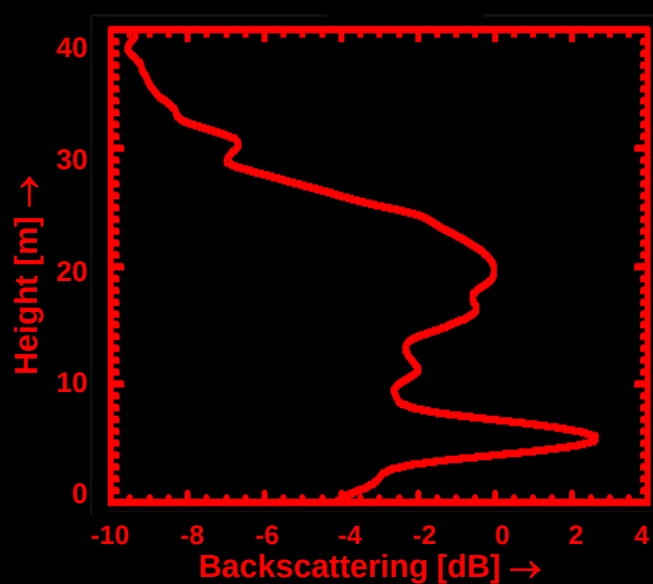
HV



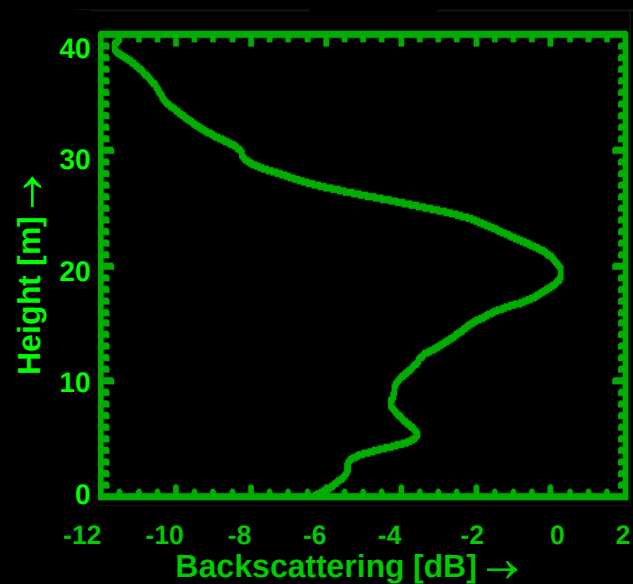
HH+VV



HH-VV

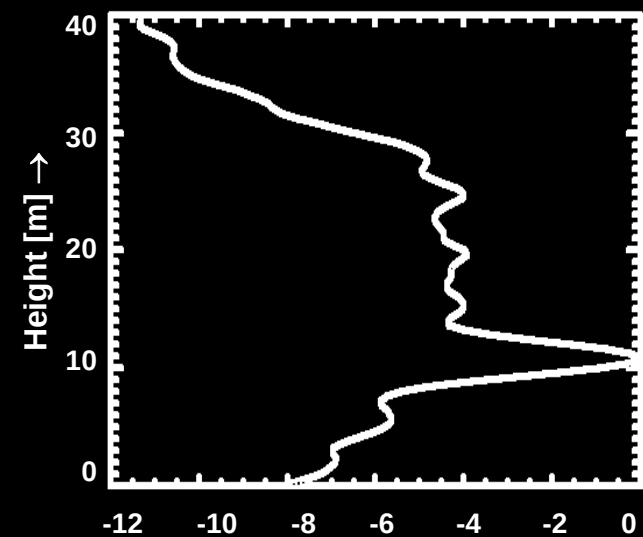


2*HV

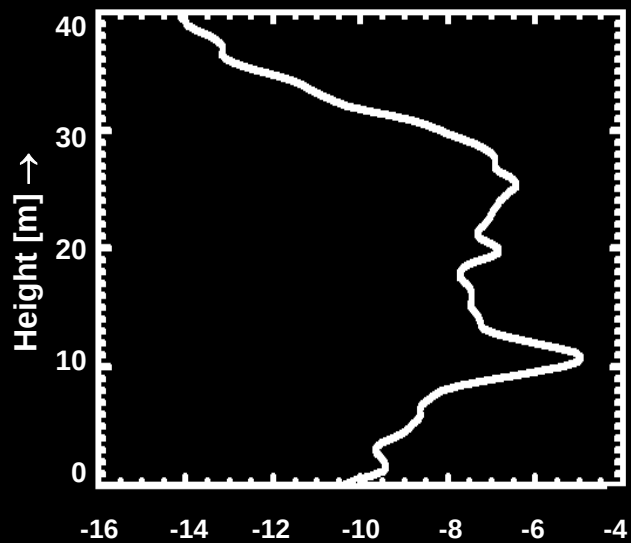


Spruce Forest Backscattering Profiles (15-20 m height)

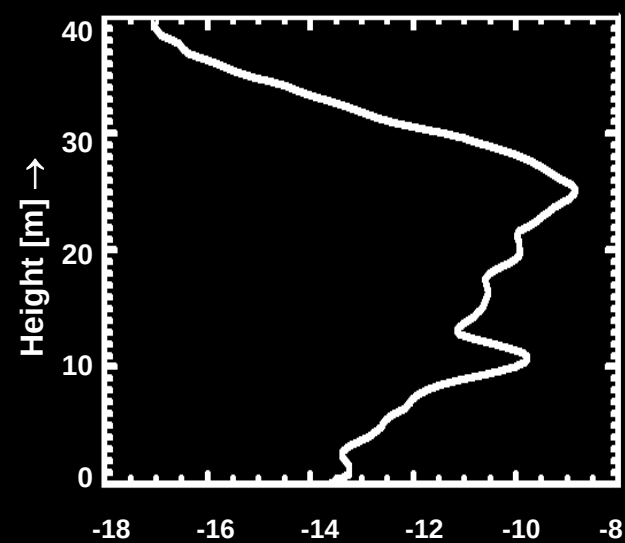
HH



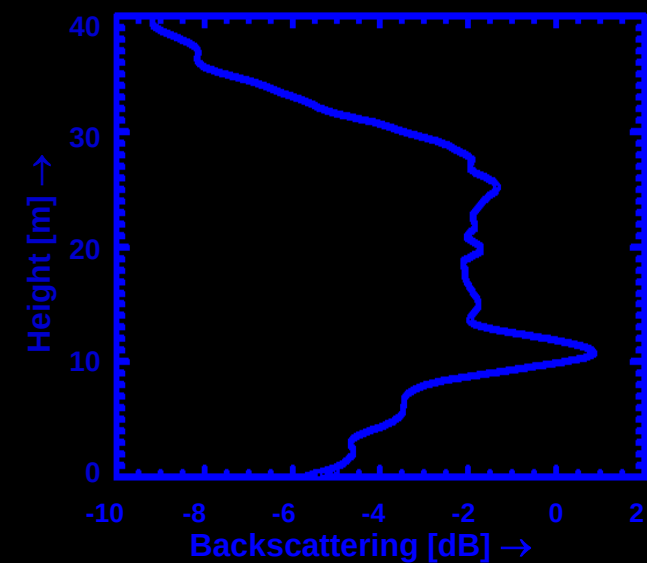
VV



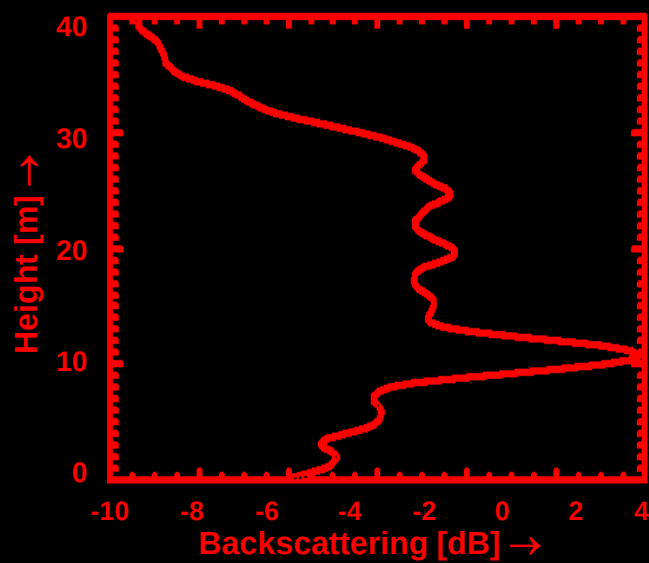
HV



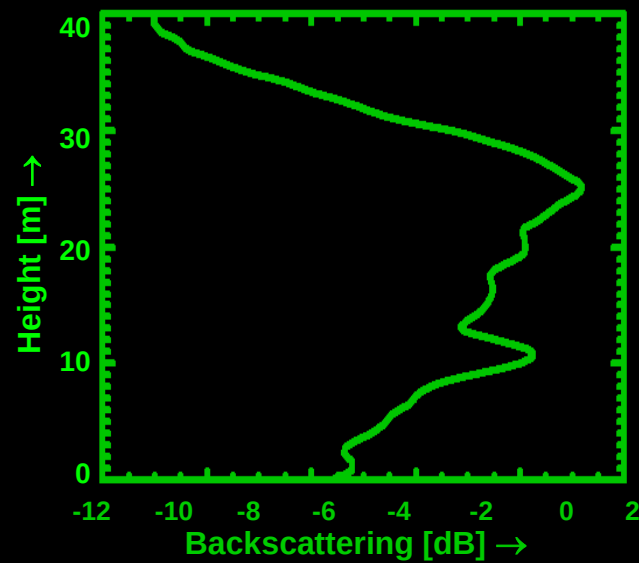
HH+VV



HH-VV



2*HV



Mixed Forest Backscattering Profiles (12-20 m height)

2 Layer Inversion Model: (Random) Volume over Ground (RVoG)



Volume Layer Ground Layer

$$f(z, \vec{w}) = m_V f_V(z) + m_G(\vec{w}) \delta(z - z_0)$$

Volume Layer Coherence

$$\tilde{Y}_V = \frac{I}{I_0} \left\{ \begin{array}{l} I = \int_0^{h_V} \exp(ik_z z') f_V(z') dz' \\ I_0 = \int_0^{h_V} f_V(z') dz' \end{array} \right.$$

$$\tilde{Y}_{Vol}(\vec{w}, \kappa_z) = \exp(i\phi_0) \frac{\tilde{Y}_V(\kappa_z) + m(\vec{w})}{1 + m(\vec{w})}$$

$$m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0} \quad \kappa_z = \frac{\kappa \Delta \theta}{\sin(\theta_0)}$$

$f_V(z)$... volume reflectivity function

$\phi_0 = k_z z_0$... underlying topography

Single Baseline Observations

single- / dual- / quad-pol

$$\tilde{Y}_{Vol}(\vec{w}_1, \kappa_z) \quad \tilde{Y}_{Vol}(\vec{w}_2, \kappa_z) \quad \tilde{Y}_{Vol}(\vec{w}_3, \kappa_z)$$

1, 2, or 3 complex coherences

Total Coherence

$$\tilde{Y}(\vec{w}, \kappa_z) = Y_{Temp}(\kappa_z) \tilde{Y}_{Vol}(\vec{w}, \kappa_z)$$

For a Single Baseline

3+N unknown parameters

Volume Height h_V

Topography ϕ_0

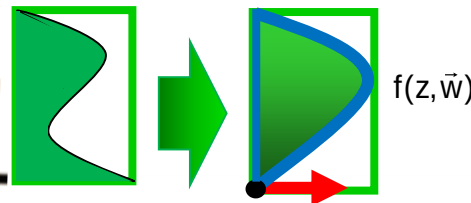
G/V Ratio $m(\vec{w}) = f(\text{pol})$

$f_V(z)$... N parameters

plus one more

Temporal Deco Y_{Temp}

for repeat-pass implementations



2 Layer Inversion Model: (Random) Volume over Ground (RVoG)



Volume Layer Ground Layer

$$f(z, \vec{w}) = m_V f_V(z) + m_G(\vec{w}) \delta(z - z_0)$$

Volume Layer Coherence

$$\tilde{Y}_V = \frac{I}{I_0} \left\{ \begin{array}{l} I = \int_0^{h_V} \exp(ik_z z') f_V(z') dz' \\ I_0 = \int_0^{h_V} f_V(z') dz' \end{array} \right. \quad m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0} \quad k_z = \frac{\kappa \Delta \theta}{\sin(\theta_0)}$$

$$\tilde{Y}_{Vol}(\vec{w}, \kappa_z) = \exp(i\phi_0) \frac{\tilde{Y}_V(\kappa_z) + m(\vec{w})}{1 + m(\vec{w})}$$

$f_V(z)$... volume reflectivity function

$\phi_0 = k_z z_0$... underlying topography

Single Baseline Observations

single- / dual- / quad-pol

$$\tilde{Y}_{Vol}(\vec{w}_1, \kappa_z) \quad \tilde{Y}_{Vol}(\vec{w}_2, \kappa_z) \quad \tilde{Y}_{Vol}(\vec{w}_3, \kappa_z)$$

1, 2, or 3 complex coherences

Total Coherence

$$\tilde{Y}(\vec{w}, \kappa_z) = \tilde{Y}_{Vol}(\vec{w}, \kappa_z)$$

For a Single Baseline

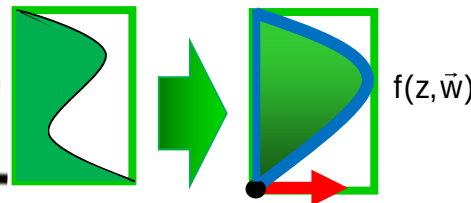
3+N unknown parameters

Volume Height h_V

Topography ϕ_0

G/V Ratio $m(\vec{w}) = f(\text{pol})$

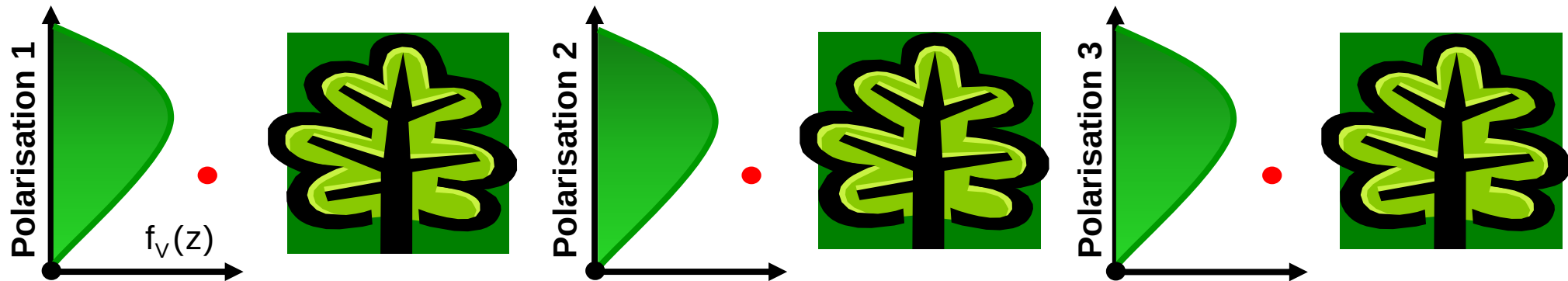
$f_V(z)$... N parameters



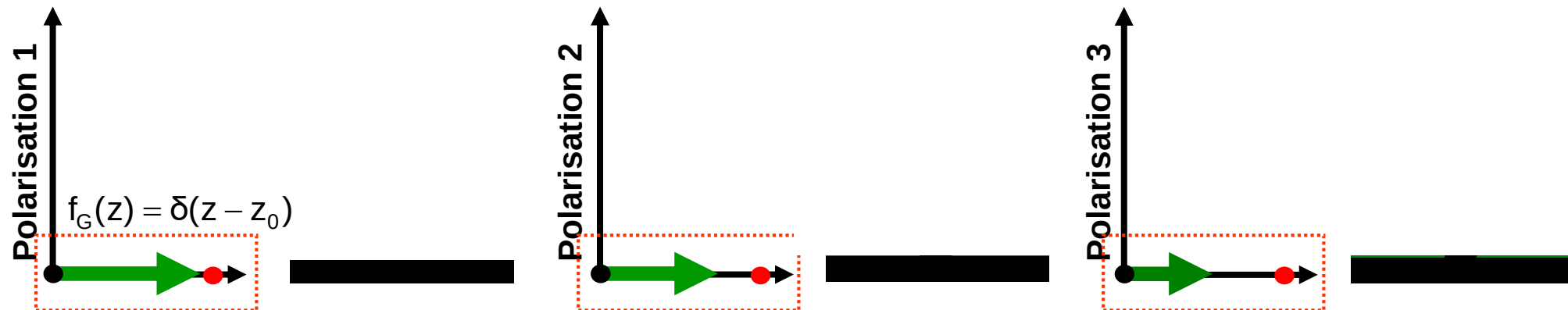
The Random Volume over Ground (RVoG) Model

The RVoG model relies on two assumptions:

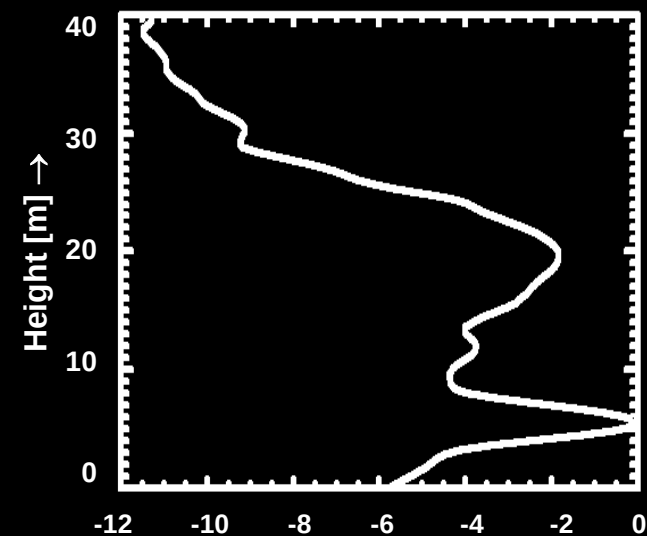
1. The vertical reflectivity function of the volume layer $f_v(z)$ is (up to a scaling factor $m_v(\vec{w})$) the same for all polarisations. All polarisations see the same (vertical) distribution of scatterers in the volume;



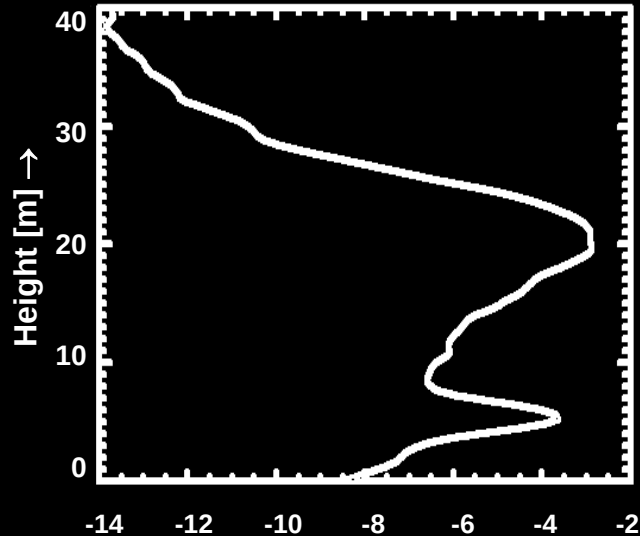
2. The Ground is impenetrable, i.e. the vertical reflectivity of the bare ground is given by a Dirac delta modulated by a (strongly) polarimetric dependent amplitude $m_G = m_G(\vec{w})$



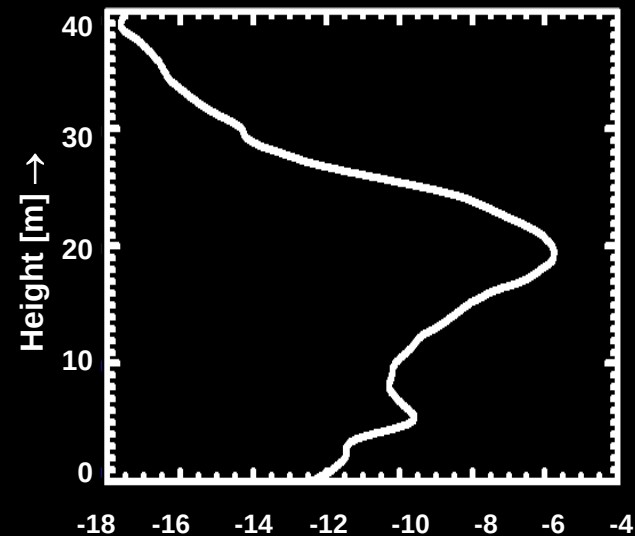
HH



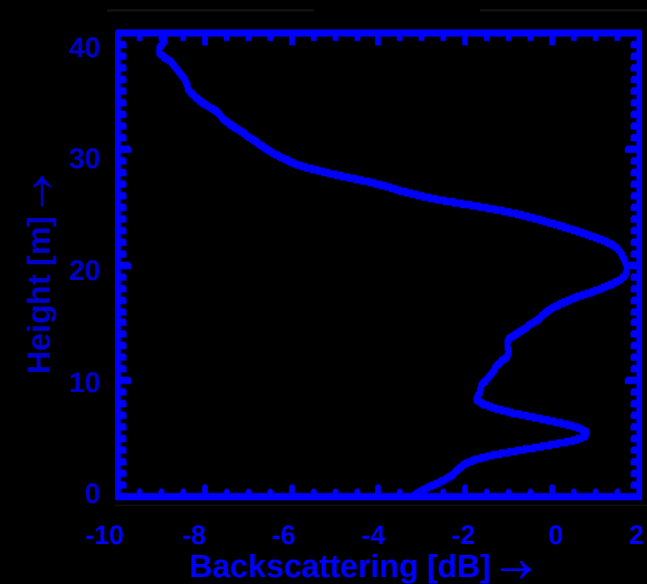
VV



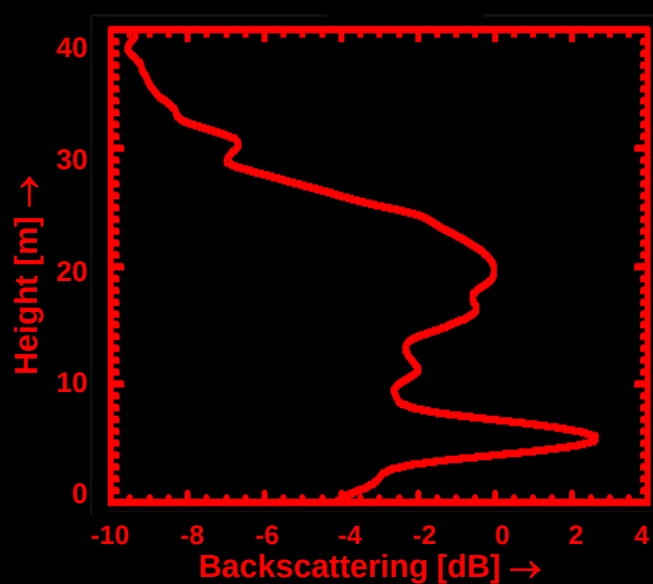
HV



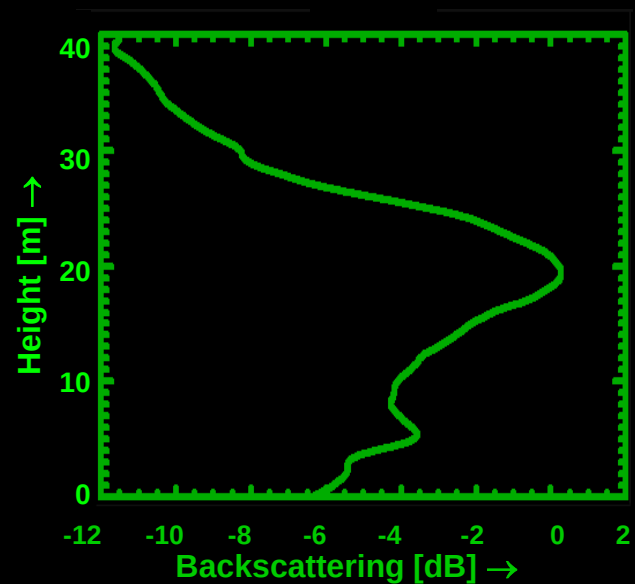
HH+VV



HH-VV



2*HV



Spruce Forest Backscattering Profiles (15-20 m height)

2 Layer Inversion Model with exponential volume reflectivity



Volume Layer Ground Layer

$$f(z, \vec{w}) = m_V f_V(z) + m_G(\vec{w}) \delta(z - z_0)$$

➔

$$\tilde{Y}_{Vol}(\vec{w}, \kappa_z) = \exp(i\phi_0) \frac{\tilde{Y}_V(\kappa_z) + m(\vec{w})}{1 + m(\vec{w})}$$

Volume Layer Coherence

$$\tilde{Y}_V = \frac{I}{I_0} \left\{ \begin{array}{l} I = \int_0^{h_V} \exp(ik_z z') e^{\left(\frac{2 \sigma z'}{\cos \theta_0}\right)} dz' \\ I_0 = \int_0^{h_V} e^{\left(\frac{2 \sigma z'}{\cos \theta_0}\right)} dz' \end{array} \right.$$

$$m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0} \quad \kappa_z = \frac{\kappa \Delta \theta}{\sin(\theta_0)}$$

$$f_V(z) = \exp\left(\frac{2 \sigma z}{\cos \theta_0}\right) \quad \text{exponential volume reflectivity}$$

Single Baseline Observations

single- / dual- / quad-pol

$\tilde{Y}_{Vol}(\vec{w}_1, \kappa_z) \quad \tilde{Y}_{Vol}(\vec{w}_2, \kappa_z) \quad \tilde{Y}_{Vol}(\vec{w}_3, \kappa_z)$

1, 2, or 3 complex coherences

Total Coherence

$$\tilde{Y}(\vec{w}, \kappa_z) = \tilde{Y}_{Temp}(\vec{w}, \kappa_z) \tilde{Y}_{Vol}(\vec{w}, \kappa_z)$$

For a Single Baseline

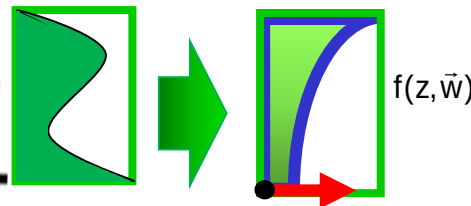
4 unknown parameters

Volume Height h_V

Topography ϕ_0

G/V Ratio $m(\vec{w}) = f(\text{pol})$

$f_V(z) \dots 1$ parameter (σ)

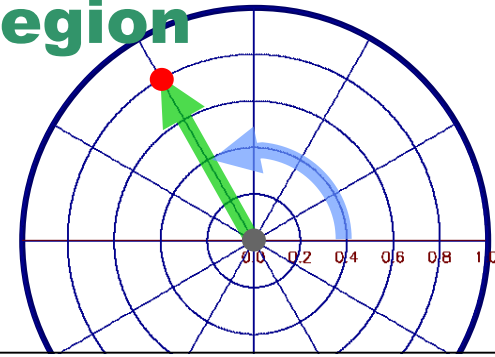


Geometrical Representation: The Coherence Region

Interferometric Coherence: $\tilde{\gamma}(\vec{w}_i, \vec{w}_i) = |\tilde{\gamma}(\vec{w}_i, \vec{w}_i)| \cdot \exp(i \text{Arg}\{\tilde{\gamma}(\vec{w}_i, \vec{w}_i)\})$

Radius

Angle



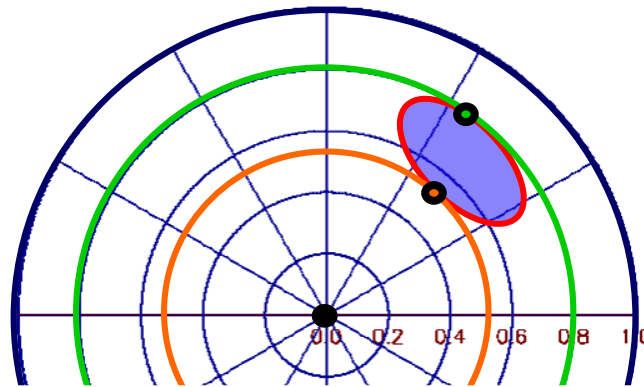
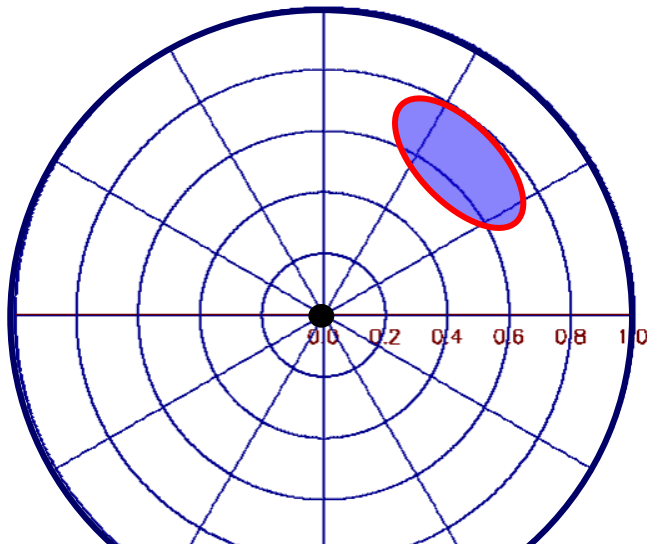
► can be represented by a single point on the unit circle (UC)

Coherence Region:

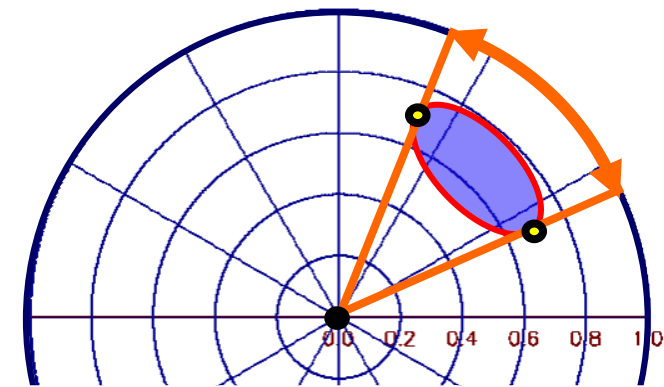
$$\tilde{\gamma}(\vec{w}_i, \vec{w}_i) \quad \forall \quad \vec{w}_i = \begin{bmatrix} \cos \alpha \exp(i\phi_1) \\ \sin \alpha \cos \beta \exp(i\phi_2) \\ \sin \alpha \sin \beta \exp(i\phi_3) \end{bmatrix} \quad \text{with} \quad 0 \leq \alpha \leq \frac{\pi}{2} \quad \text{and} \quad -\pi \leq \beta \leq \pi$$

The set of interferometric coherences obtained for all the possible polarizations $\vec{w}_i$ plotted on the unit circle (UC) defines the so-called **coherence region**.

Its shape & size depend on the structure of the underlying scatterer and on acquisition parameters.



Max./ Min. Interferometric Coherence
as function of $\vec{w}_i$



Max. Phase Difference between
interferograms formed with $\vec{w}_i$ and $\vec{w}_j$

RVoG Geometrical Interpretation

Interferometric RVoG formulation (for a single spatial baseline given by the vertical wavenumber κ_z):

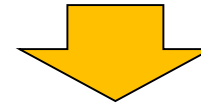
Volume Ground

$$f(z, \vec{w}) = m_v f_v(z) + m_G(\vec{w}) \delta(z - z_0)$$

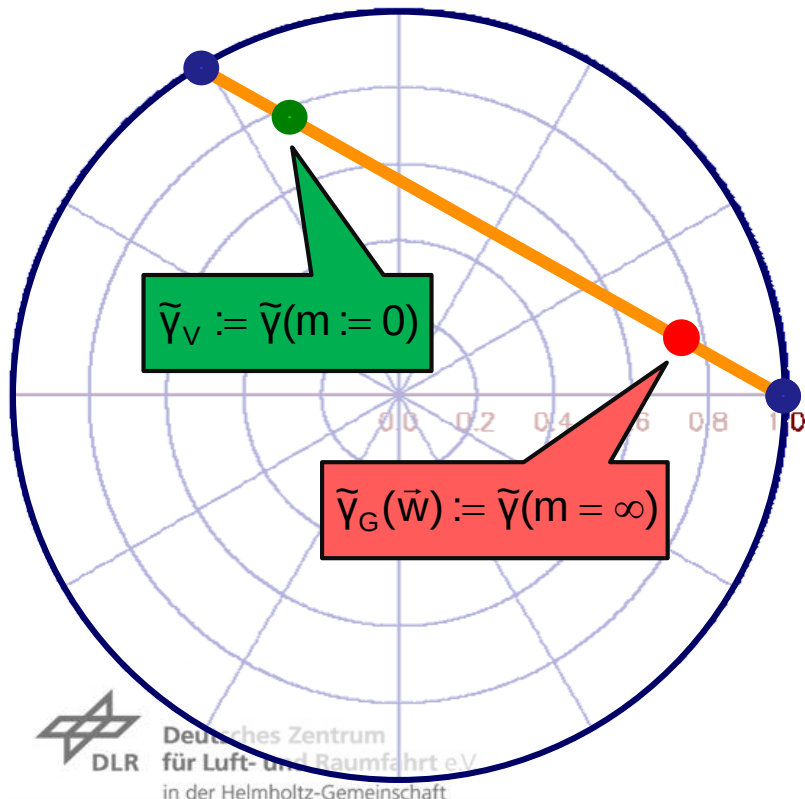


$$\tilde{Y}_{Vol}(\vec{w}, \kappa_z) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}) dz}$$

$$\tilde{Y}_{Vol}(\vec{w}, \kappa_z) = e^{ik_z z_0} \frac{\tilde{Y}_V(\kappa_z) + m(\vec{w})}{1 + m(\vec{w})}$$



$$\tilde{Y}_{Vol}(\vec{w}) = \exp(ik_z z_0) [\tilde{Y}_V + F(\vec{w})(1 - \tilde{Y}_V)] \quad \text{where} \quad F(\vec{w}) = \frac{m(\vec{w})}{1 + m(\vec{w})}$$



The RVoG solution space is a line segment inside the UC:

The (complex) interferometric coherence $\tilde{Y}_{Vol}(\vec{w})$ vary as a function of polarisation (i.e. Ground-to-Volume ratio m) along a straight line segment inside the unit circle:

- Interferometric coherence of the volume: $\tilde{Y}(m := 0) = \tilde{Y}_V$
- Interferometric coherence of the ground: $\tilde{Y}(m = \infty) = \tilde{Y}_G(\vec{w})$

In the absence of ground decorrelation $\tilde{Y}_G(\vec{w})$ lies on the UC !



RVoG Coherence Region

Image 1: $[S_1] = \begin{bmatrix} S_{HH}^1 & S_{HV}^1 \\ S_{VH}^1 & S_{VV}^1 \end{bmatrix}$



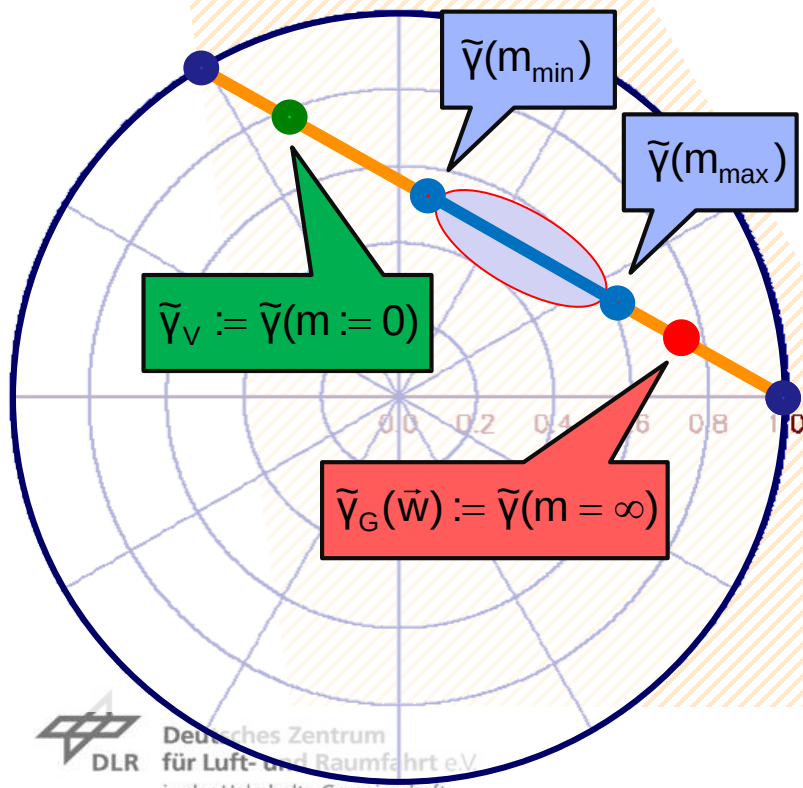
$$\vec{k}_1 = \frac{1}{\sqrt{2}} [S_{HH}^1 + S_{VV}^1 \quad S_{HH}^1 - S_{VV}^1 \quad 2S_{HV}^1]^T$$

Image 2: $[S_2] = \begin{bmatrix} S_{HH}^2 & S_{HV}^2 \\ S_{VH}^2 & S_{VV}^2 \end{bmatrix}$



$$\vec{k}_2 = \frac{1}{\sqrt{2}} [S_{HH}^2 + S_{VV}^2 \quad S_{HH}^2 - S_{VV}^2 \quad 2S_{HV}^2]^T$$

$$\left\{ \begin{array}{l} [T_{11}] = \langle \vec{k}_1 \cdot \vec{k}_1^+ \rangle \\ [T_{22}] = \langle \vec{k}_2 \cdot \vec{k}_2^+ \rangle \\ [\Omega_{12}] = \langle \vec{k}_1 \cdot \vec{k}_2^+ \rangle \end{array} \right.$$

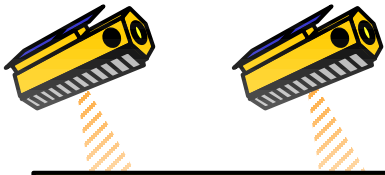


**Interferometric
Coherence**

$$\tilde{\gamma}(\vec{w}) = \frac{\langle \vec{w}^+ [\Omega_{12}] \vec{w} \rangle}{\sqrt{\langle \vec{w}^+ [T_{11}] \vec{w} \rangle \langle \vec{w}^+ [T_{22}] \vec{w} \rangle}}$$

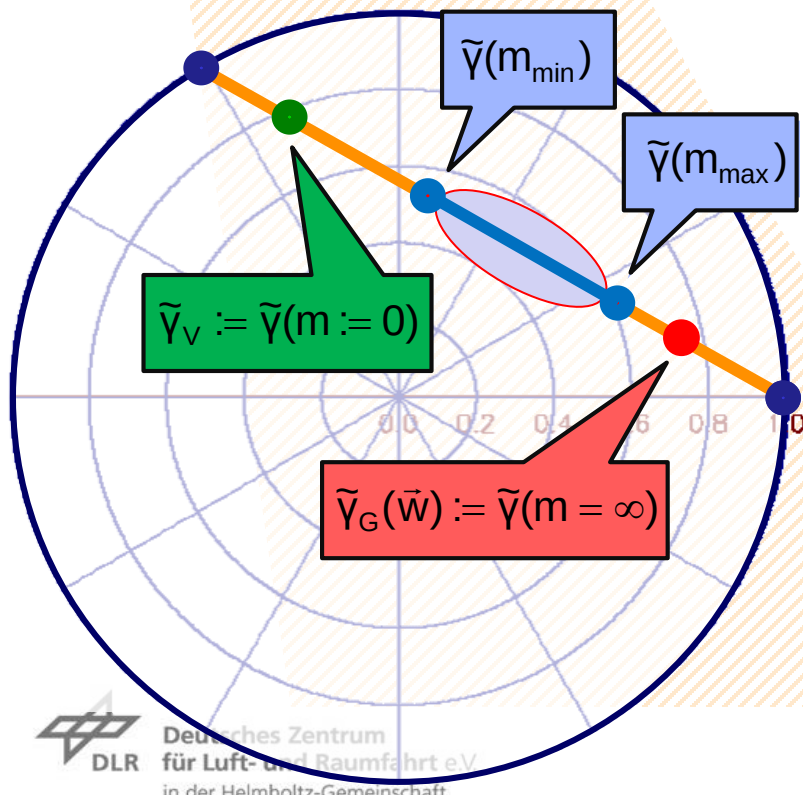
The **Coherence Region** is a line segment inside the Unit Circle limited by $\tilde{\gamma}(m_{max})$ and $\tilde{\gamma}(m_{min})$ and is only a subspace of the RVoG solution space.

The interferometric coherences associated to the ground $\tilde{\gamma}(m = \infty) = \tilde{\gamma}_G(\vec{w})$ and volume $\tilde{\gamma}(m := 0) = \tilde{\gamma}_V$ are (in general) not within the Coherence Region and therefore ambiguous.



RVoG Coherence Region

- To establish unique solutions regularisation conditions are required.
- Their choice is critical as it impacts on obtained estimates and/or the achieved performance.
- Their performance strongly depends on the boundary conditions of the individual inversion problem.



**Interferometric
Coherence**

$$\tilde{\gamma}(\vec{w}) = \frac{\langle \vec{w}^+ [\Omega_{12}] \vec{w} \rangle}{\sqrt{\langle \vec{w}^+ [T_{11}] \vec{w} \rangle \langle \vec{w}^+ [T_{22}] \vec{w} \rangle}}$$

The **Coherence Region** is a line segment inside the Unit Circle limited by $\tilde{\gamma}(m_{\max})$ and $\tilde{\gamma}(m_{\min})$ and is only a subspace of the RVoG solution space.

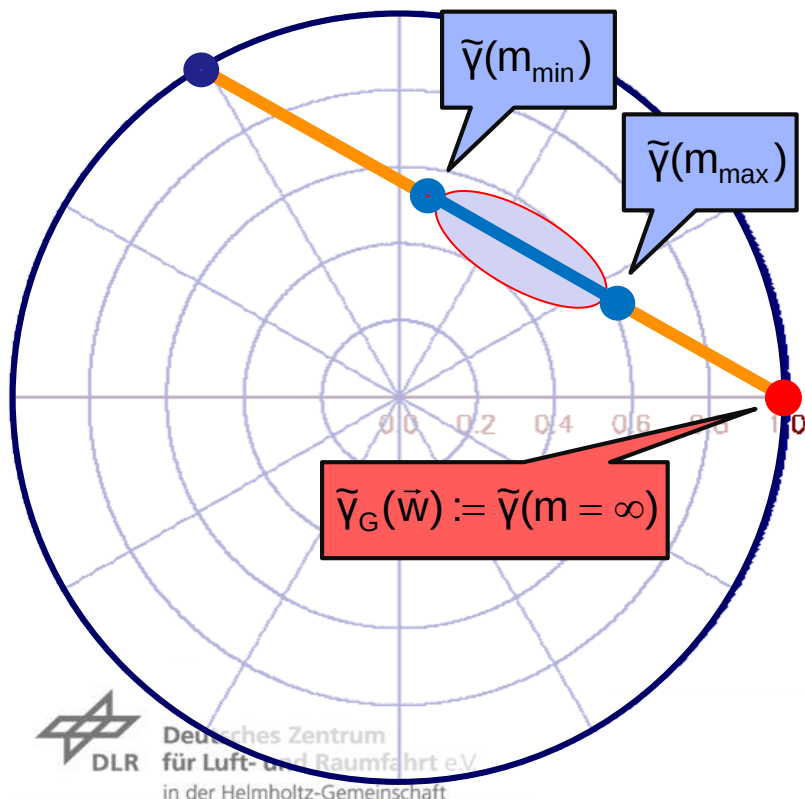
The interferometric coherences associated to the ground $\tilde{\gamma}(m = \infty) = \tilde{\gamma}_G(\vec{w})$ and volume $\tilde{\gamma}(m := 0) = \tilde{\gamma}_V$ are (in general) not within the Coherence Region and therefore ambiguous.

Establishing Uniqueness: The "Ground" Regularisation

1) The "Ground" Regularisation: the ground coherence $\tilde{\gamma}_G := \tilde{\gamma}(m = \infty)$ is known (or fixed) !!!

This is the optimum solution possible if:

- there is no (or negligible) decorrelation of the ground so that $|\tilde{\gamma}_G| := 1$ holds and
- the phase of the ground can be either estimated from the data itself or is known.



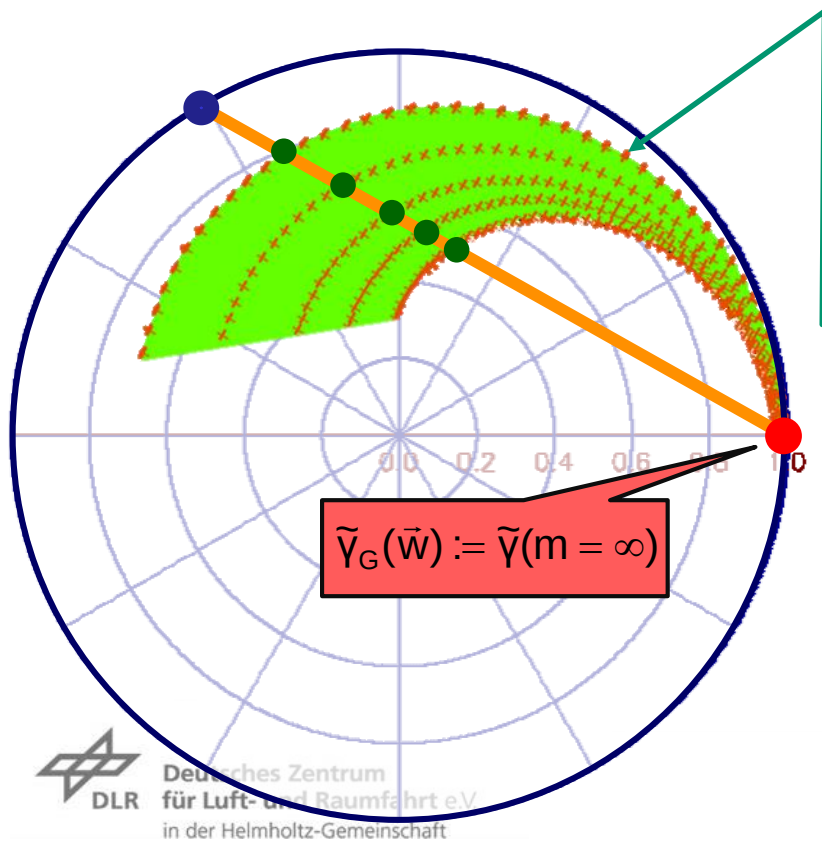
Fixing the ground coherence $\tilde{\gamma}_G = e^{ik_z z_0}$ is still not sufficient to constrain the interferometric coherence of the volume $\tilde{\gamma}(m := 0) = \tilde{\gamma}_v$ by means of a single- baseline:

Establishing Uniqueness: The "Ground" Regularisation

1) The "Ground" Regularisation: the ground coherence $\tilde{\gamma}_G := \tilde{\gamma}(m = \infty)$ is known (or fixed) !!!

This is the optimum solution possible if:

- there is no (or negligible) decorrelation of the ground so that $|\tilde{\gamma}_G| := 1$ holds and
- the phase of the ground can be either estimated from the data itself or is known.



Fixing the ground coherence $\tilde{\gamma}_G = e^{ik_z z_0}$ is still not sufficient to constrain the interferometric coherence of the volume $\tilde{\gamma}(m := 0) = \tilde{\gamma}_V$ by means of a single- baseline:

$$\tilde{\gamma}_V = \frac{\int_0^{h_V} \exp(ik_z z) f_V(z) dz}{\int_0^{h_V} f_V(z) dz}$$

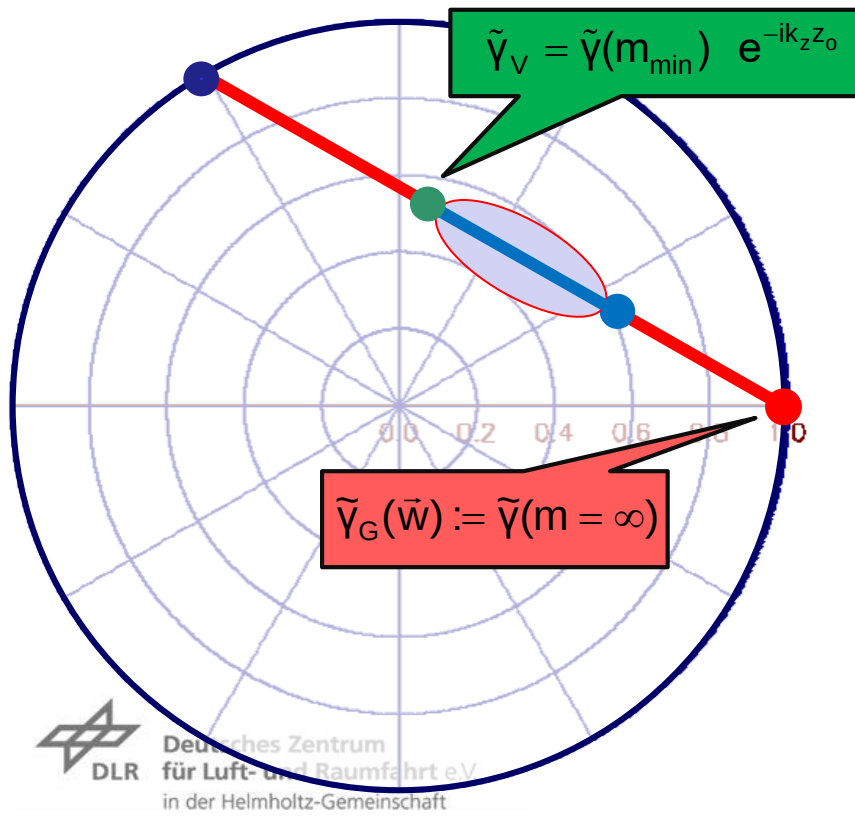
Every (volume) reflectivity function $f_V(z)$ leads for a certain height h_V to another intersection

with the RVoG solution space (i.e. with the line) and thus to an ambiguous $\tilde{\gamma}_V$ estimate !!!

Establishing Uniqueness: The “Volume” Regularisation

Unique forest height estimates require the (unique) estimation of the interf. coherence of the volume component $\tilde{\gamma}_V$ (in addition to the knowledge of the ground coherence).

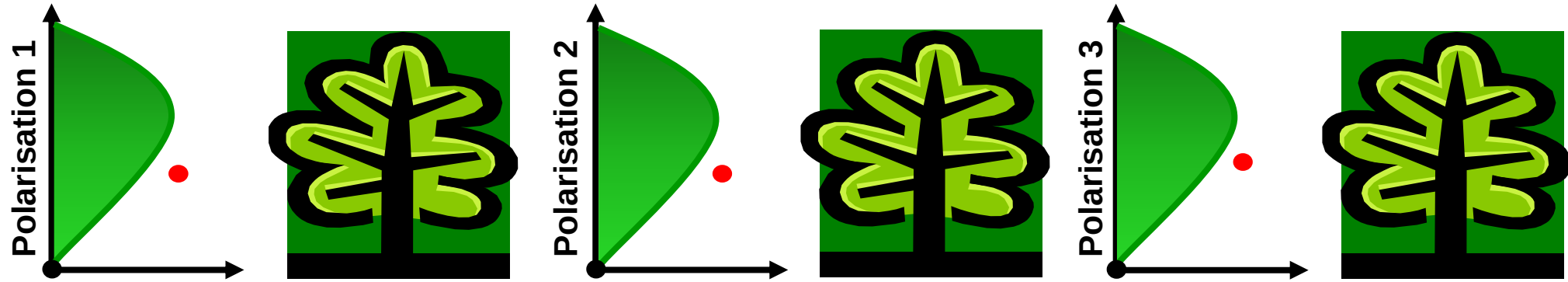
1. For a single-baseline and in the absence of temporal decorrelation the unique estimation of the interf. coherence of the volume component can be forced by assuming a two dimensional (polarimetrically) ground scattering component: $m_{\min} = 0 \rightarrow \tilde{\gamma}_V = \tilde{\gamma}(m_{\min}) e^{-ik_z z_0}$



Note that this is equivalent to “constraining” the volume reflectivity function $f_V(z)$ in order to allow only height solutions h_V that intersect the “line” at $\tilde{\gamma}(m_{\min})$!

2. Introducing a second baseline allows to obtain unique estimates of the interferometric coherence of the volume component without the need to require a two dimensional ground scattering. Instead, a parameterisation of the volume reflectivity function $f_V(z)$ has to be provided.

Polarimetric Behaviour: Random vs. Oriented Volume

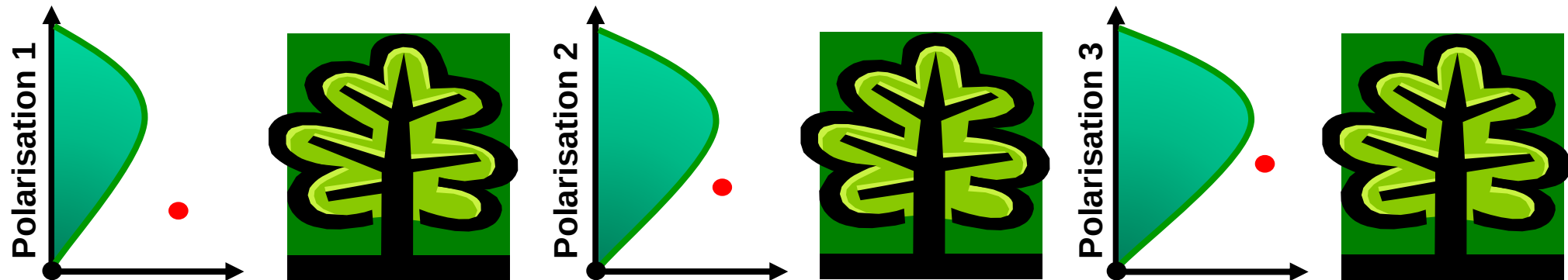


Random Volume: The vertical reflectivity function is independent of polarisation (or each polarisation sees the same volume vertical reflectivity $f_v(z)$)

$$f_v := f_v(z) \mapsto \tilde{\gamma}_v(\kappa_z)$$

Oriented Volume: The vertical reflectivity function changes with polarisation (or each polarisation sees a different volume vertical reflectivity $f_v(z)$)

$$f_v := f_v(z, \vec{w}) \mapsto \tilde{\gamma}_v(\kappa_z, \vec{w})$$



Polarimetric Behaviour: 3-dim vs 2-dim Ground Scatterer

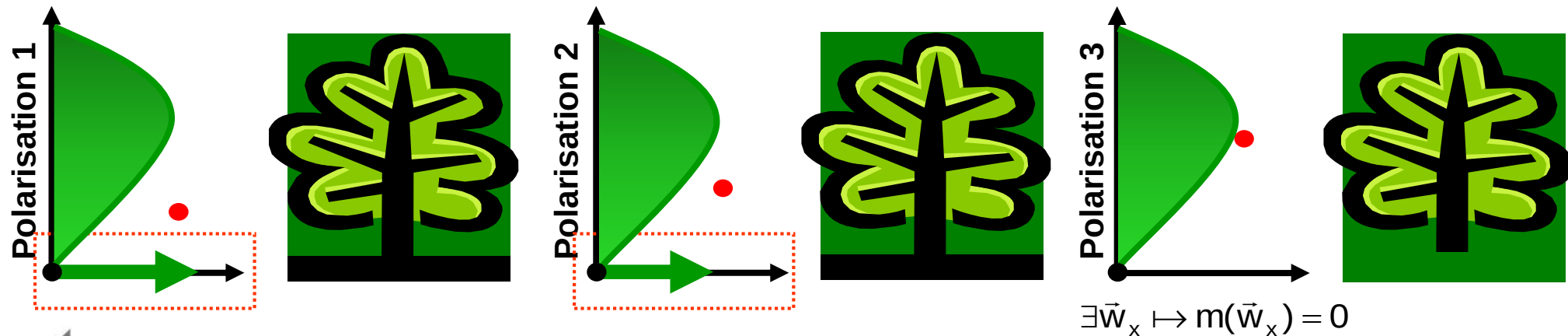


3-dim ground scatterer: A ground scattering component is visible in all polarisations (or there is no polarisation that “switches-off” the ground)

$$\forall \vec{w} \quad m(\vec{w}) \neq 0$$

2-dim ground scatterer: There is (at least) one polarisation in which the ground disappears

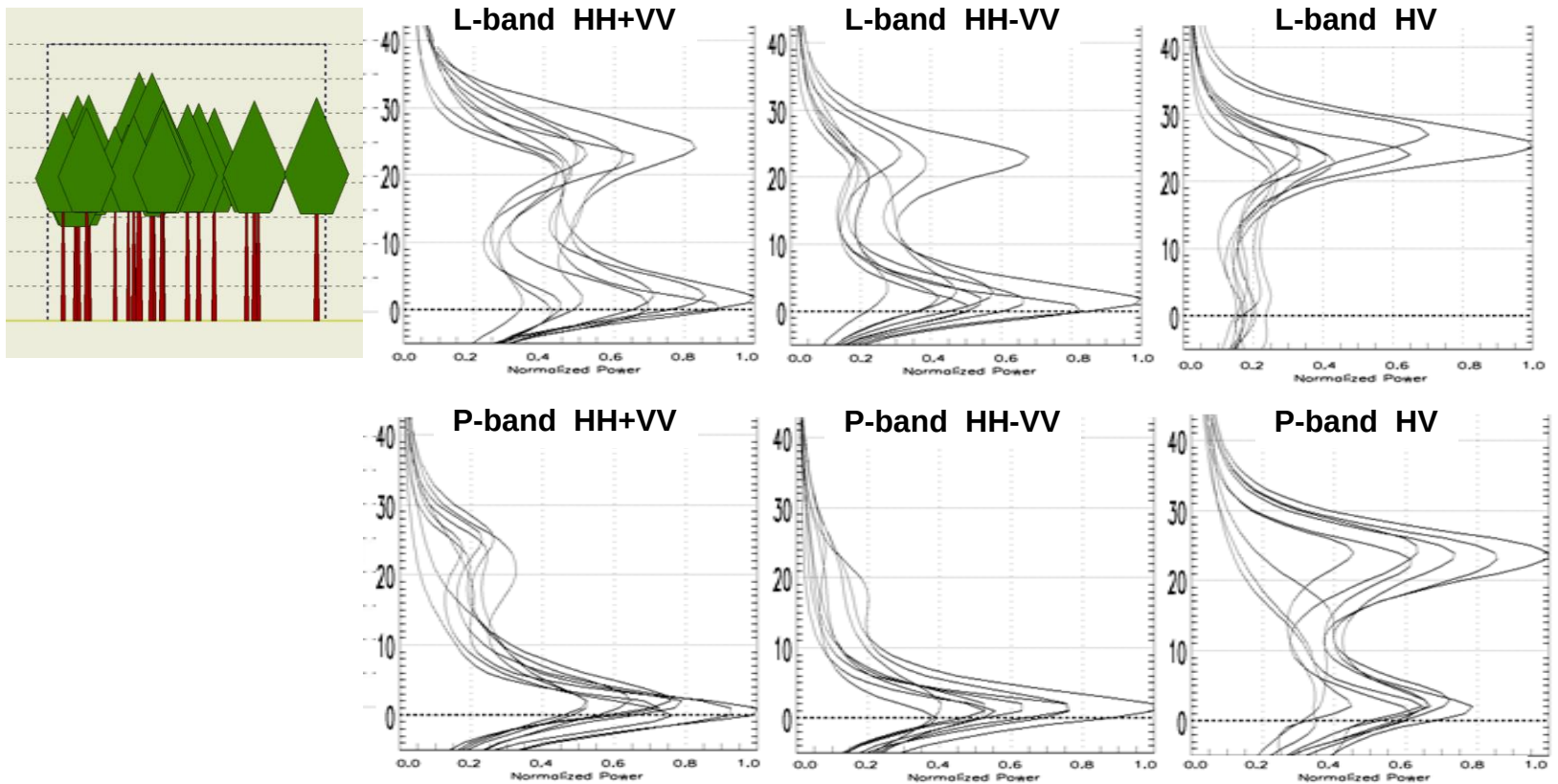
$$\exists \vec{w} \mapsto m(\vec{w}) = 0$$



$$\exists \vec{w}_x \mapsto m(\vec{w}_x) = 0$$

Vertical Reflectivity Profiles: L- and P-band

Reflectivity Profiles (6 Tracks L-band)

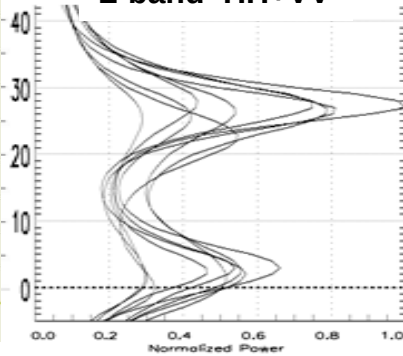


Vertical Reflectivity Profiles: L- and P-band

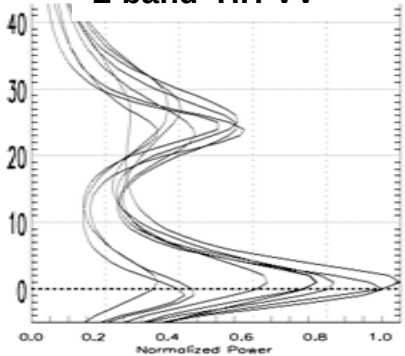
Reflectivity Profiles (6 Tracks L-band)



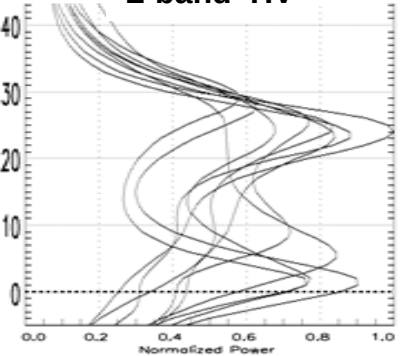
L-band HH+VV



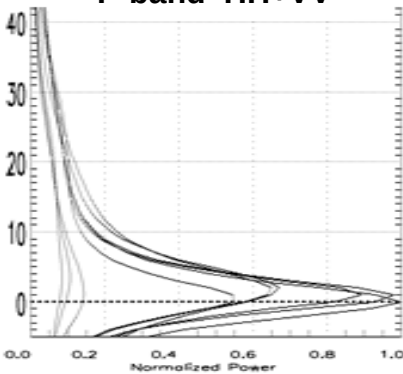
L-band HH-VV



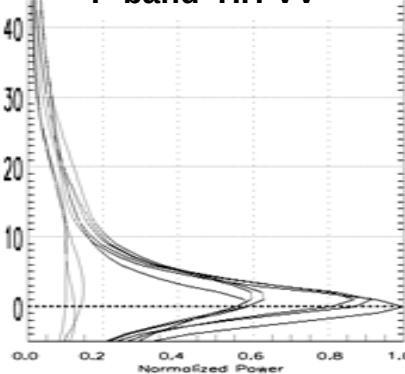
L-band HV



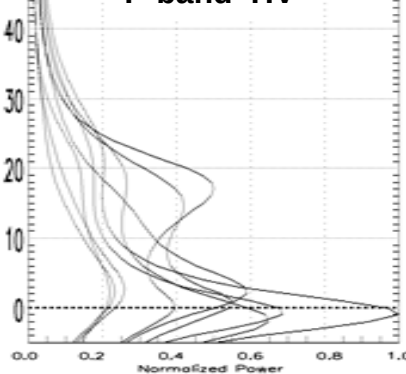
P-band HH+VV



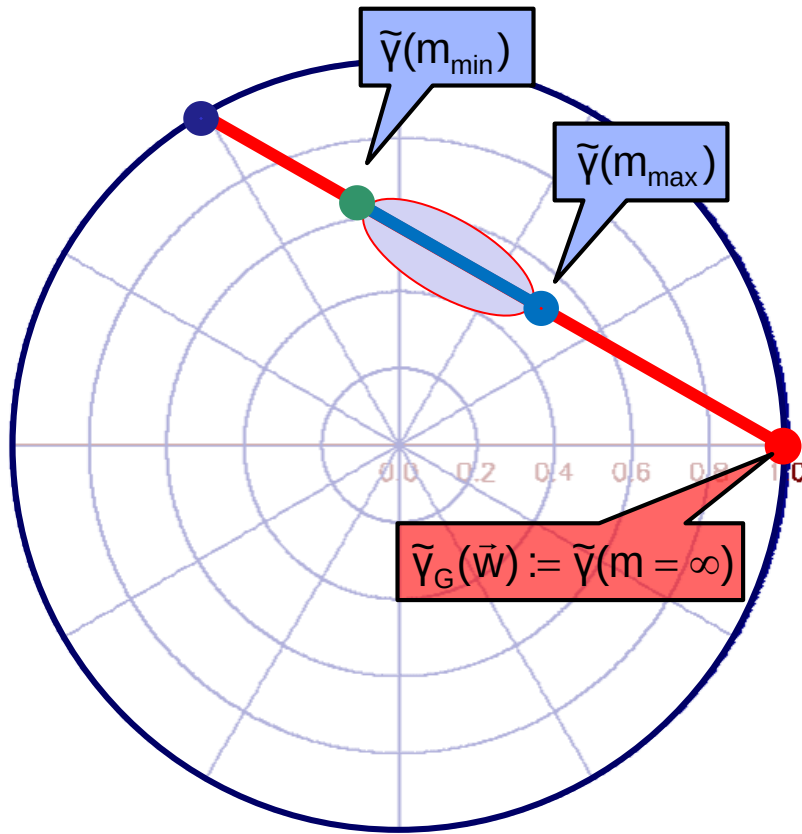
P-band HH-VV



P-band HV



Single Baseline RVoG Solution: Geometric Representation

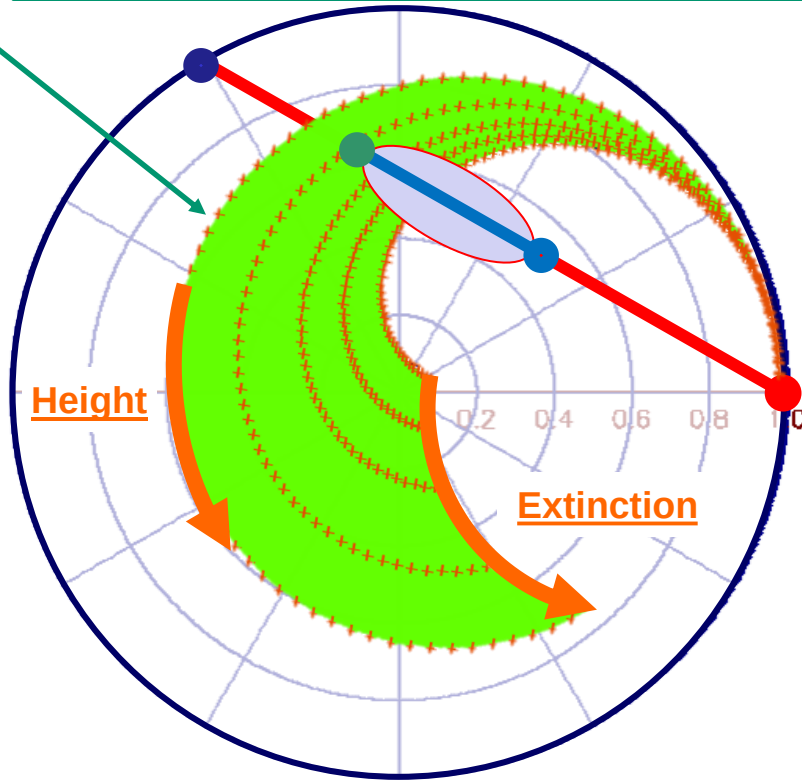


Inversion Recipe

1. Estimation of the Coherence Region (CR);
2. Line fit through the extreme points of the CR
 $\tilde{\gamma}(m_{\min})$ and $\tilde{\gamma}(m_{\max})$;
3. Estimation of the line-circle intersection point corresponding to the underlying ground $\tilde{\gamma}(m = \infty) = \exp(i\phi_0)$

Single Baseline RVoG Solution: Geometric Representation

Curve of constant extinction σ and variable height h_v



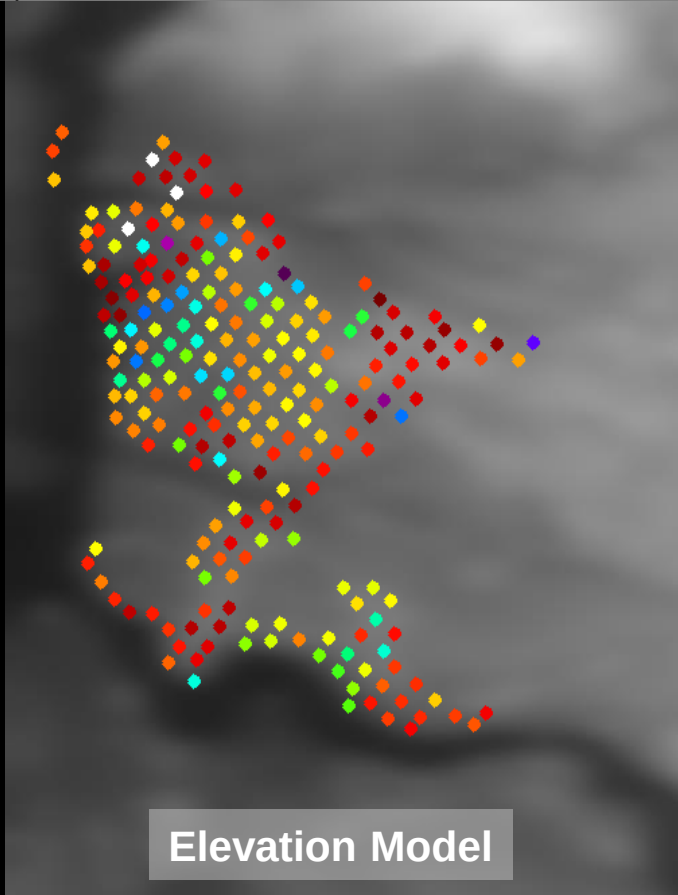
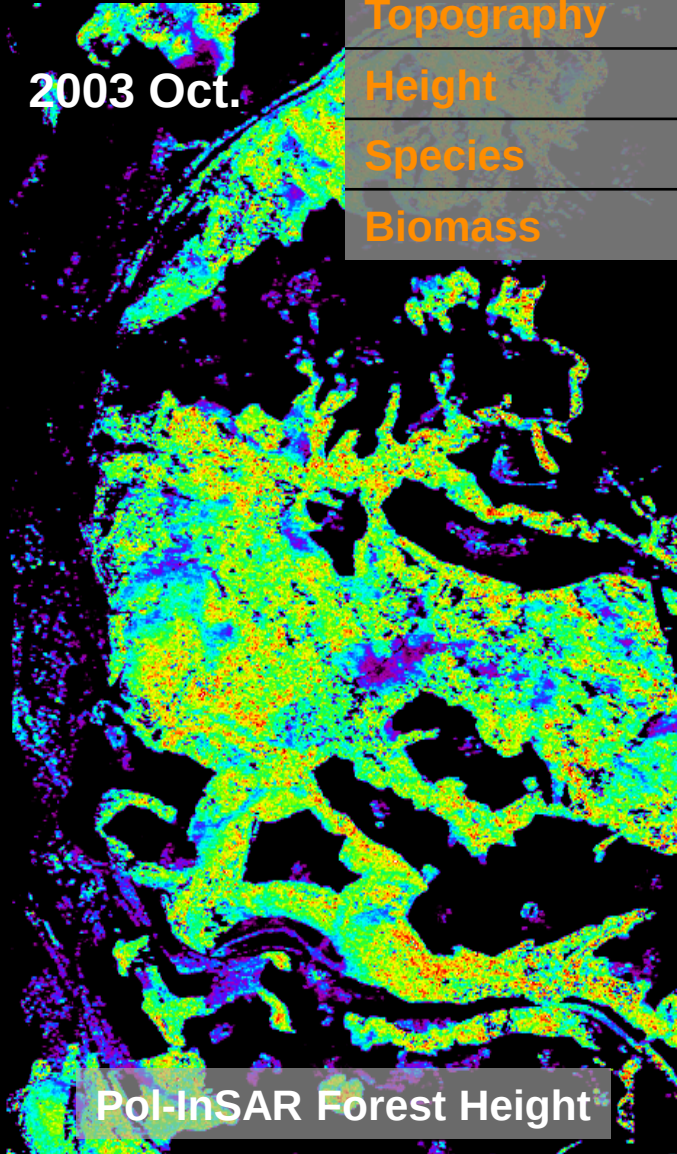
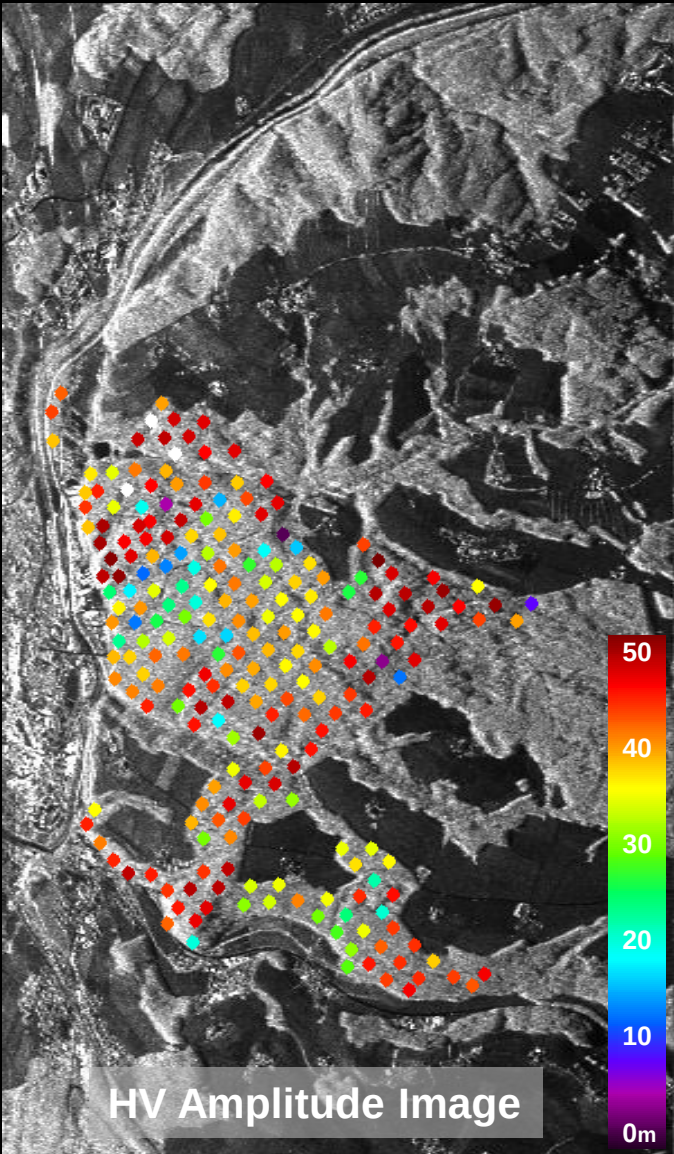
Inversion Recipe

1. Estimation of the Coherence Region (CR);
2. Line fit through the extreme points of the CR
 $\tilde{\gamma}(m_{\min})$ and $\tilde{\gamma}(m_{\max})$;
3. Estimation of the line-circle intersection point corresponding to the underlying ground $\tilde{\gamma}(m = \infty) = \exp(i\varphi_0)$
4. From the underlying ground point $\tilde{\gamma}(m = \infty) = \exp(i\varphi_0)$ a Volume Height–Extinction Look-Up Table (LUT) is initialised that provides at every intersection with the line a {extinction σ , height h_v } solution couple.
5. The intersection of $\tilde{\gamma}(m_{\min})$ with the LUT provides the right {extinction σ , height h_v } solution couple

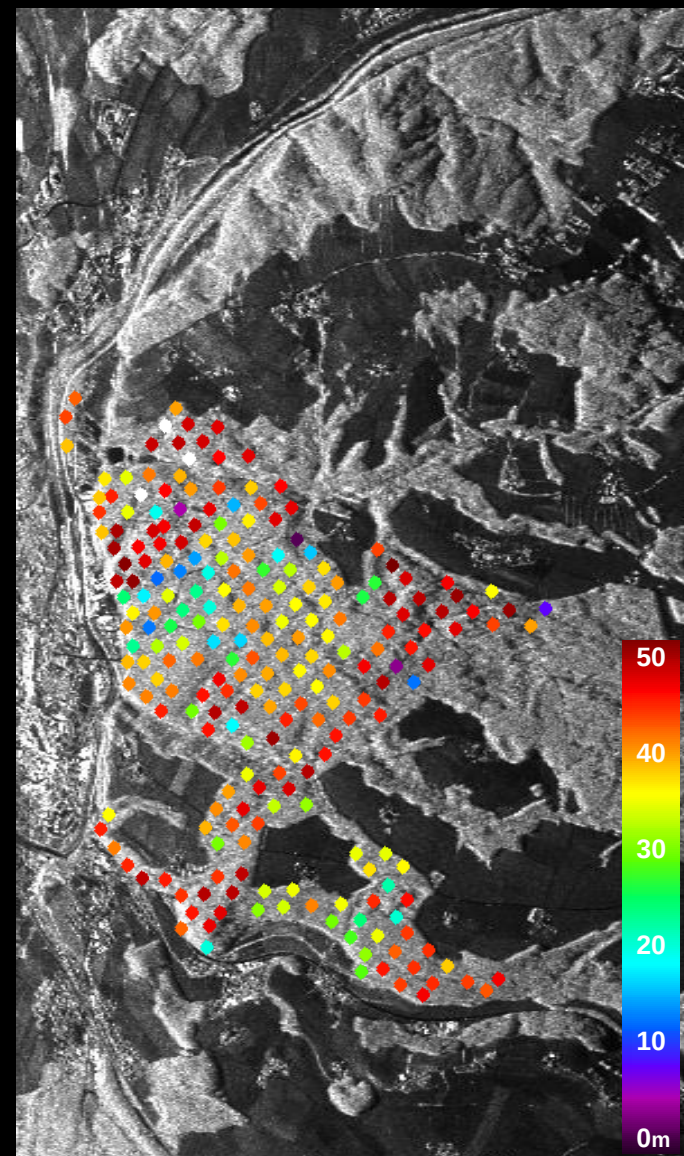
$$\tilde{\gamma}(m = 0) = \exp(i\varphi_0) \frac{\int_0^{h_v} \exp(ik_z z') \exp\left(\frac{2 \sigma z'}{\cos \theta_0}\right) dz'}{\int_0^{h_v} \exp\left(\frac{2 \sigma z'}{\cos \theta_0}\right) dz'}$$

Traunstein Test Site

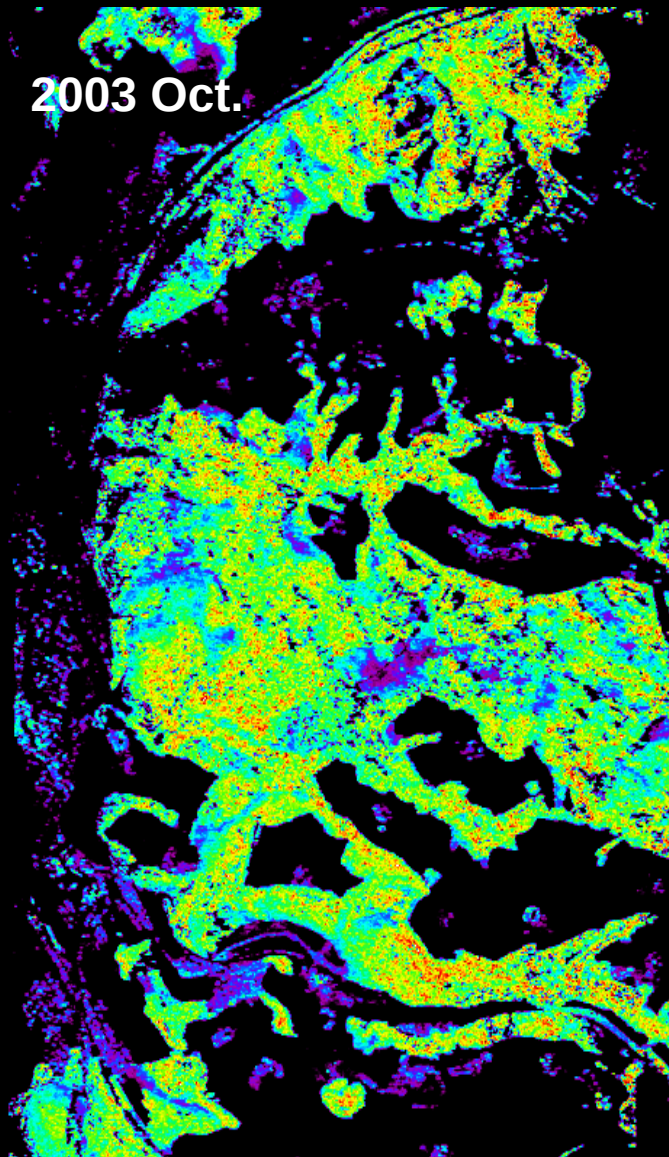
Forest type	Temperate
Topography	Moderate slopes
Height	25 ~ 35m
Species	N. Spruce, E. Beech, White Fir
Biomass	40 ~ 450 t/ha



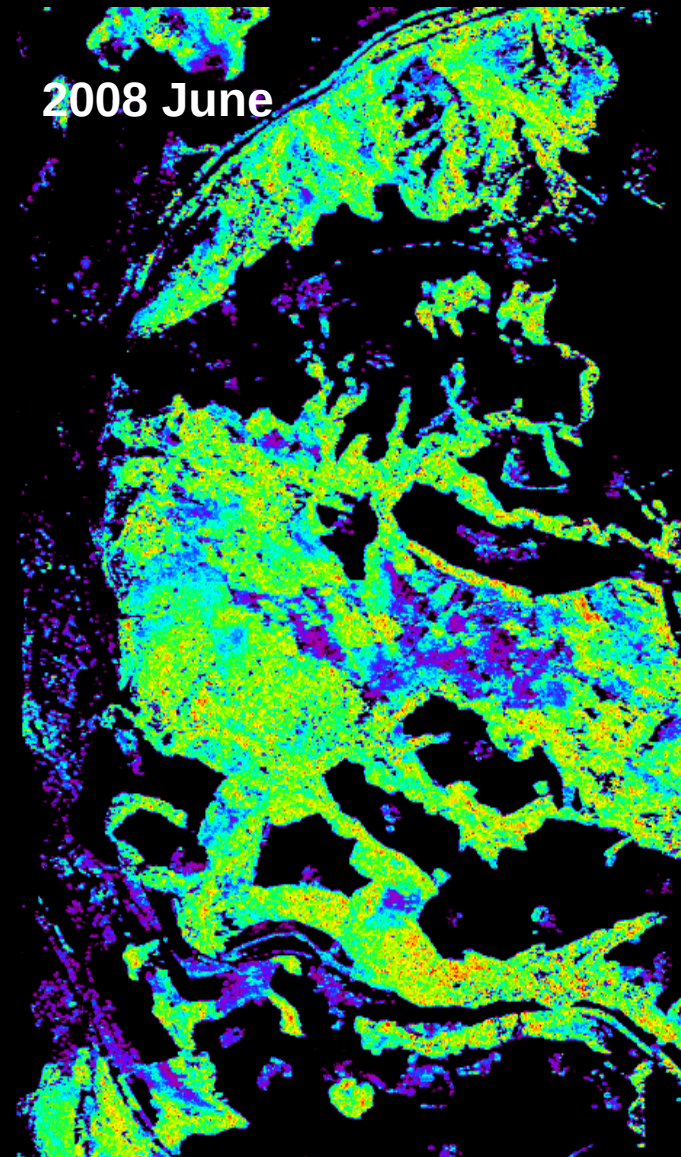
Traunstein Test Site



2003 Oct.



2008 June



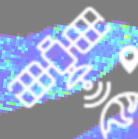
The Course

Part 9: Pol-InSAR 2

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

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ESAMAAP

EEBI  **MASS**

Polarimetric SAR Interferometry

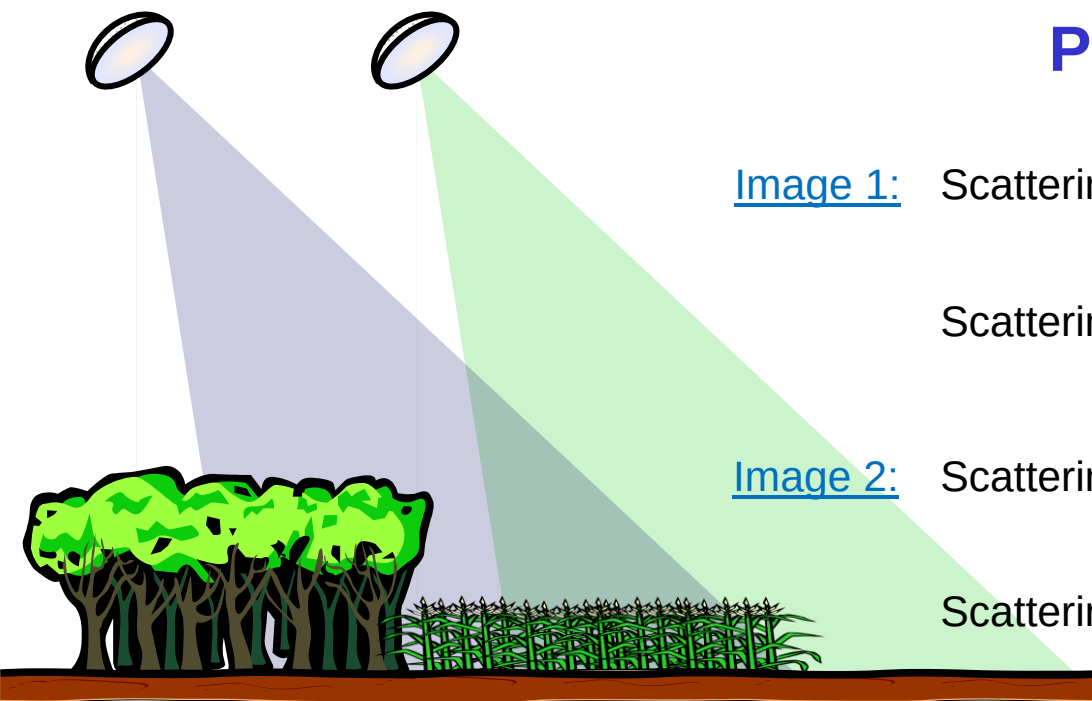


Image 1: Scattering Matrix: $[S_1] = \begin{bmatrix} S_{HH}^1 & S_{HV}^1 \\ S_{VH}^1 & S_{VV}^1 \end{bmatrix}$

Scattering Vector: $\vec{k}_1 = \frac{1}{\sqrt{2}} [S_{HH}^1 + S_{VV}^1 \quad S_{HH}^1 - S_{VV}^1 \quad 2S_{HV}^1]^T$

Image 2: Scattering Matrix: $[S_2] = \begin{bmatrix} S_{HH}^2 & S_{HV}^2 \\ S_{VH}^2 & S_{VV}^2 \end{bmatrix}$

Scattering Vector: $\vec{k}_2 = \frac{1}{\sqrt{2}} [S_{HH}^2 + S_{VV}^2 \quad S_{HH}^2 - S_{VV}^2 \quad 2S_{HV}^2]^T$

1. Pol-InSAR Matrices: The Dual-Pol Case

A: Polarisation 1, B: Polarisation 2

- Two scattering vectors $\vec{k}_1$ and $\vec{k}_2$ from $[S_1]$ and $[S_2]$: $\vec{k}_1 = \frac{1}{\sqrt{2}} [S_A^1 \quad S_B^1]^T$ and $\vec{k}_2 = \frac{1}{\sqrt{2}} [S_A^2 \quad S_B^2]^T$
- Polarimetric coherency matrices: $[T_1] = \langle \vec{k}_1 \cdot \vec{k}_1^{*T} \rangle = \begin{bmatrix} S_A^1 S_A^{1*} & S_A^1 S_B^{1*} \\ S_B^1 S_A^{1*} & S_B^1 S_B^{1*} \end{bmatrix}$, $[T_2] = \langle \vec{k}_2 \cdot \vec{k}_2^{*T} \rangle = \begin{bmatrix} S_A^2 S_A^{2*} & S_A^2 S_B^{2*} \\ S_B^2 S_A^{2*} & S_B^2 S_B^{2*} \end{bmatrix}$
- Polarimetric interferometric matrix: $[\Omega] = \langle \vec{k}_1 \cdot \vec{k}_2^{*T} \rangle = \begin{bmatrix} S_A^1 S_A^{2*} & S_A^1 S_B^{2*} \\ S_B^1 S_A^{2*} & S_B^1 S_B^{2*} \end{bmatrix}$

Pol-InSAR Matrices: The Dual-Pol Case

$$[T_1] = \langle \vec{k}_1 \cdot \vec{k}_1^{*T} \rangle, \quad [T_2] = \langle \vec{k}_2 \cdot \vec{k}_2^{*T} \rangle, \quad [\Omega] = \langle \vec{k}_1 \cdot \vec{k}_2^{*T} \rangle$$

Pre-Whitening and Normalisation

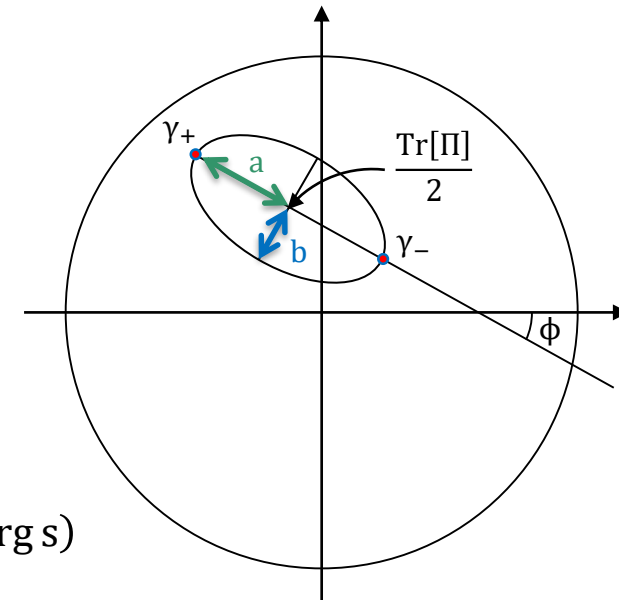
- Normalization (Pre-whitening): $[\Pi] = [T]^{-\frac{1}{2}}[\Omega][T]^{-\frac{1}{2}}$ with $[T] = \frac{1}{2}([T_1] + [T_2])$ and $[T]^{\frac{1}{2}}[T]^{\frac{1}{2}} = [T]$:

$$[\Pi] = \begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{bmatrix} \quad \text{with} \quad \text{Tr}([\Pi]) = \Pi_{11} + \Pi_{22}.$$

- Interferometric coherence from $[\Pi]$: $\tilde{\gamma}(\vec{w}_1, \vec{w}_2) = \langle \vec{w}_1^+ [\Pi] \vec{w}_2 \rangle$

Geometrical Representation: The Coherence Region

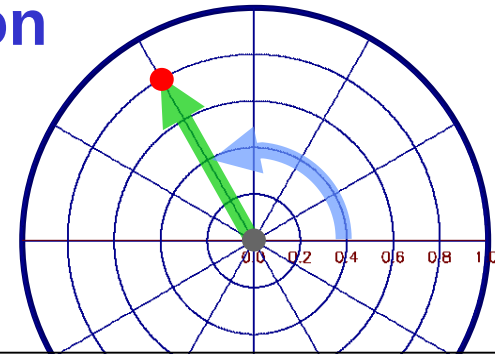
- Major axis: $2a = |r| + |s|$
- Minor axis: $2b = ||r| - |s||$
- Center: $c = \frac{\text{Tr}[\Pi]}{2}$
- Orientation (direction of major axis w.r.t. the x-axis): $\phi = \frac{1}{2}(\arg r + \arg s)$
- Vertices of $[\Pi]$: $\gamma_{\pm} = \frac{\text{Tr}[\Pi]}{2} \pm ae^{i\phi}$



Coherence Optimization: The Polar Decomposition

Interferometric Coherence: $\tilde{\gamma}(\vec{w}_i, \vec{w}_i) = \underbrace{|\tilde{\gamma}(\vec{w}_i, \vec{w}_i)|}_{\text{Radius}} \cdot \exp(i \underbrace{\text{Arg}\{\tilde{\gamma}(\vec{w}_i, \vec{w}_i)\}}_{\text{Angle}})$

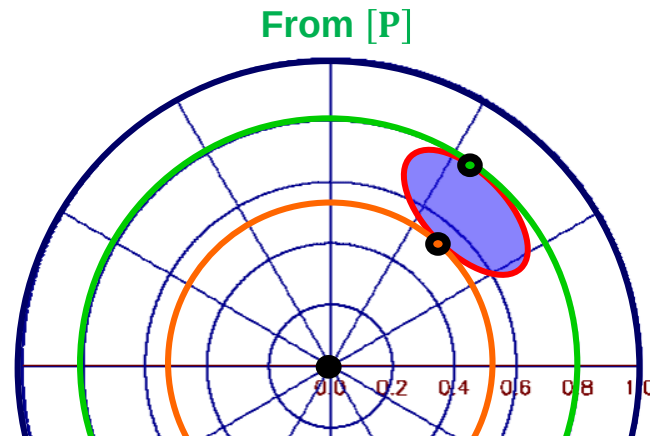
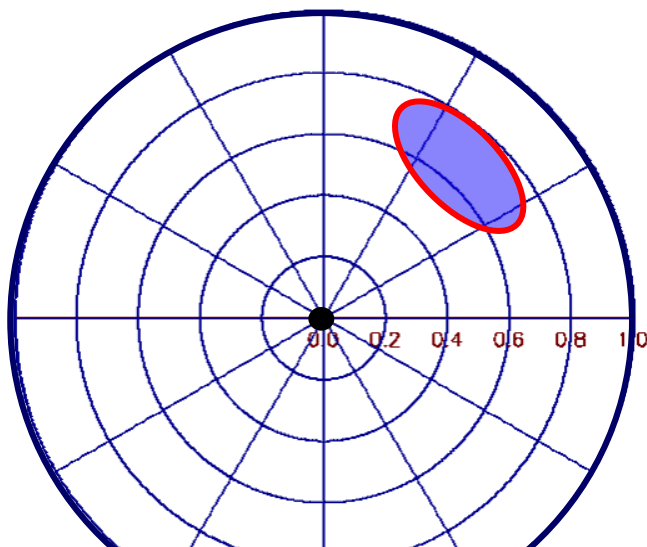
► can be represented in matrix notation with the Polar Decomposition.



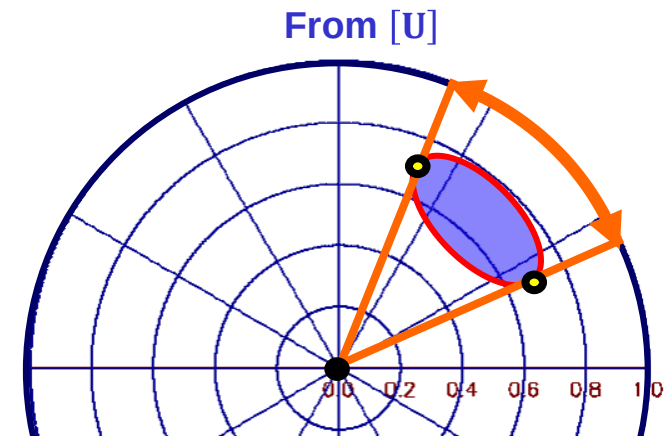
Polar Decomposition: $[\Pi] = [U] \cdot [P]$, where $\begin{cases} [P] = ([\Pi]^*{}^T [\Pi])^{\frac{1}{2}} & \text{(positive semi-definite Hermitian matrix)} \\ [U] = [\Pi][P]^{-1} & \text{(unitary matrix)} \end{cases}$

► **Amplitude** can be maximized/minimized by the eigenvectors of $[P]$: $\gamma_{\max}, \gamma_{\min}$

► **Phase** can be maximized/minimized by the eigenvectors of $[U]$: $\gamma(\phi_{\min}), \gamma(\phi_{\min})$



Max./ Min. Interferometric Coherence
as function of $\vec{w}_i$



Max. Phase Difference between
interferograms formed with $\vec{w}_i$ and $\vec{w}_j$

PolInSAR Inversion

1. Estimate the volume-only coherence:

- For a positive (negative) vertical wavenumber, the coherence with the larger (smaller) angle indicates a higher elevation: $\tilde{\gamma}(m_{\min}) \equiv \tilde{\gamma}_{V0}$, used to estimate h_v .
- The coherence with the smaller angle, $\tilde{\gamma}(m_{\max})$, is closest to the topographic phase $e^{i\phi_0}$.

2. Calculate the ground phase:

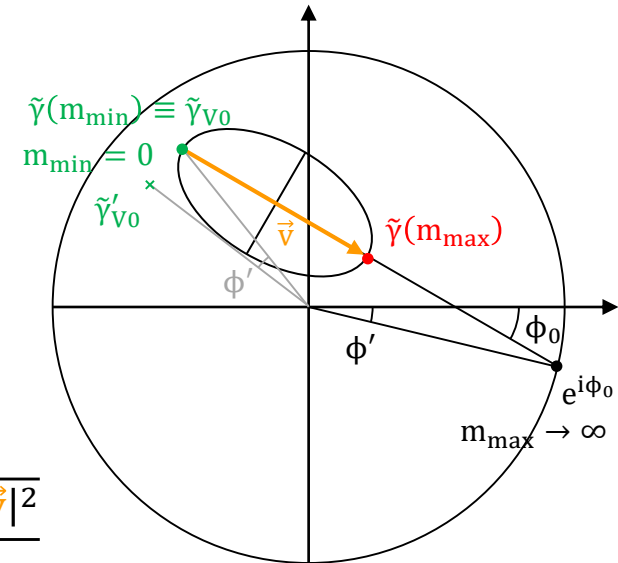
- Set displacement: $\vec{v} = \tilde{\gamma}(m_{\max}) - \tilde{\gamma}_{V0}$
- For a positive real number x , it is satisfied:

$$(\tilde{\gamma}_{V0} + \vec{v}x)(\tilde{\gamma}_{V0} + \vec{v}x)^* = 1$$

- Solving quadratic equation yields

$$x = -\frac{\Re\{\vec{v}^* \tilde{\gamma}_{V0}\}}{|\vec{v}|^2} + \frac{\sqrt{(\Re\{\vec{v}^* \tilde{\gamma}_{V0}\})^2 - (|\tilde{\gamma}_{V0}|^2 - 1)|\vec{v}|^2}}{|\vec{v}|^2}$$

- The phase of $\tilde{\gamma}_{V0} + \vec{v}x$ is the ground phase, or $e^{i\phi_0} = \tilde{\gamma}_{V0} + \vec{v}x$



PolInSAR Inversion

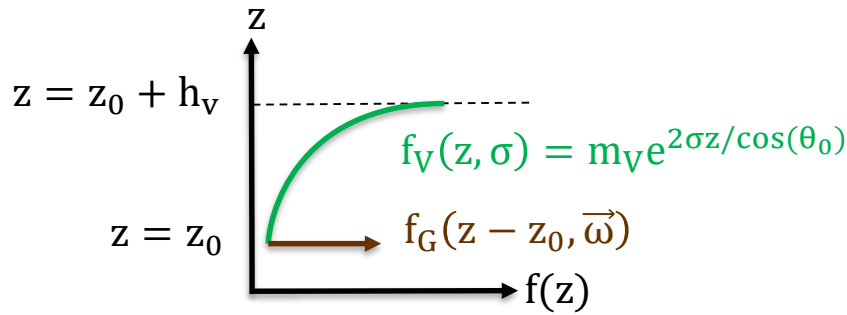
3. Compensate the ground phase:

➤ Pol-InSAR forest height inversion is performed for rotated volume phase: $\tilde{\gamma}'_{V0} = \tilde{\gamma}_{V0} e^{-i\phi_0}$

4. Create Look-Up Table (LUT):

RVoG

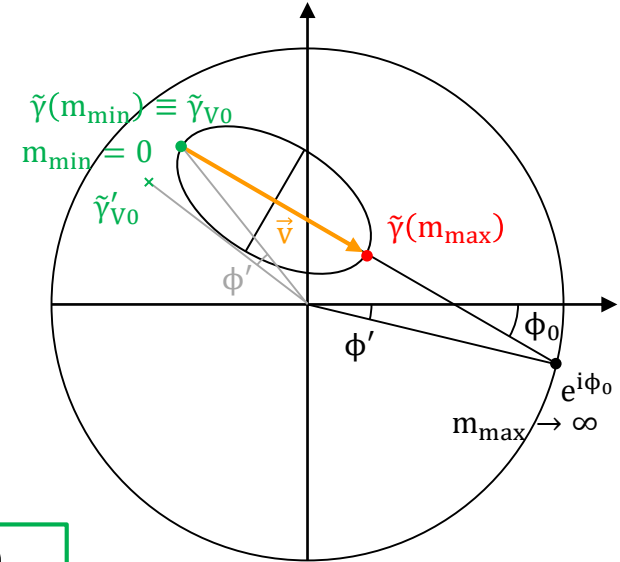
$$F(z, \vec{\omega}) = f_V(z, \vec{\omega}) + m_G(\vec{\omega})\delta(z - z_0)$$



$$\tilde{\gamma}_{Vol}(m = 0) = e^{i\phi_0} \frac{\int_0^{h_v} f_V(z) e^{i\kappa_Z z} dz}{\int_0^{h_v} f_V(z) dz}$$

$$\tilde{\gamma}_V = f(h_v, \sigma) = e^{i\kappa_Z z_0} \frac{p_1 (e^{p_2 h_v} - 1)}{p_2 (e^{p_1 h_v} - 1)},$$

where $\begin{cases} p_1 = \frac{2\sigma}{\cos(\theta_0)} \\ p_2 = \frac{2\sigma}{\cos(\theta_0)} + i\kappa_Z \end{cases}$



- If $h_v = 0$: $\tilde{\gamma}_V = 1$
- If $\sigma = 0$, $m_{\min} = 0$:

$$\tilde{\gamma}_V = e^{i\frac{\kappa_Z h_v}{2}} \text{sinc}\left(\frac{\kappa_Z h_v}{2}\right)$$

5. Estimate and validate height: $\tilde{\gamma}_V$ from LUT $\rightarrow h_v, \sigma$

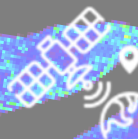
The Course

Part 9: Pol-InSAR 2

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